Chapter 25

Resin-bonded Splints and Bridges

"If you cannot do great things, do small things in a great way…"

—Napolean Hill

Mobility of teeth has many causes, including traumatic injury to the face, advanced periodontal disease, habits such as thumb sucking and tongue thrusting and malocclusion. In addition, teeth often need stabilization and retention after orthodontic treatment. In the past, clinical procedures for the stabilization of teeth either involved extensive loss of the tooth structure or were poor in appearance. A conservative and esthetic alternative has been made possible by using resin-bonded splints.

Certain criteria must be met when mobile teeth are splinted. Occlusal adjustment may be necessary initially. The splint should have a hygienic design so that the patient is able to maintain good oral hygiene. It also should allow further diagnostic procedures and treatment, if necessary. The resin-bonded splinting technique satisfies these criteria. Light-cured composites are recommended for splinting because they afford extended working time for placement and contouring.

Periodontally Involved Teeth

Loss of bone support allows movement of teeth, resulting in an increased irritation to the supporting tissues and possible malpositioning of teeth. Stabilizing mobile teeth is a valuable treatment aid before, during and after periodontal therapy. Splinting of teeth aids in occlusal adjustment and tissue healing, thus allowing better evaluation of the progression and prognosis of treatment.

A resin-bonded splint via the acid-etch technique is a conservative and effective method of protecting teeth from further injury by stabilizing them in a favourable occlusal relationship. If the periodontal problem is complicated by missing teeth, a bridge incorporating a splint design is indicated (see section Conservative Bridges).

Techniques for Splinting Anterior Teeth

In short-span segments subject to minimal occlusal forces, a relatively simple technique can be used for splinting periodontally involved teeth. Fig. 25.1A illustrates a maxillary lateral incisor that remains mobile because of insufficient bone support even after occlusal adjustment and elimination of a periodontal pocket. Esthetic recontouring with composite augmentation can be accomplished along with the splinting procedure.

- Anaesthesia generally is not required for a splinting procedure when enamel covers the clinical crown. When root surfaces are exposed and extreme sensitivity exists, however, local anaesthesia is necessary. Teeth are cleaned with a pumice slurry, and the shade of light-cured composite is selected. A cotton roll and retraction cords are used for isolation in this instance.
- With a *coarse, flame-shaped diamond instrument*, enamel on both teeth at the proximal contact area is reduced to produce an interdental space approximately **0.5-mm wide**. This amount of space enhances the strength of the splint by providing more bulk of composite material in the connector between teeth. Other enamel areas of the tooth or teeth that need more contour are prepared by roughening the surface with a coarse diamond instrument.
- Where no enamel is present, such as on the root surface, a mechanical lock is prepared with a No. ¼ round bur in the dentin at the gingivo-axial line angle of the preparation.
- After the prepared tooth surfaces are acid-etched, rinsed and dried, a lightly frosted appearance should be observed on the etched enamel surfaces (Fig. 25.1B).
- The adhesive is applied, lightly blown with air and polymerized. A hand instrument is used to place a small amount of composite material in the gingival area. Additional shaping with a No. 2 explorer reduces the amount of finishing necessary later. It is helpful to add and cure composite in small increments, building from the gingival aspect towards the incisal aspect.



Figure 25.1 Splinting and recontouring a mobile tooth using a light-cured composite. (A) Maxillary right lateral incisor is mobile from lack of bone support. (B) Preparations completed and etched. (C) Splinted and recontoured tooth after 4 years.

- Finishing is accomplished with round and flame-shaped carbide burs, fine diamonds and polishing disks and points. The retraction cord is removed and the occlusion is evaluated to assess centric contacts and functional movements. Instructions on brushing and flossing are reviewed with the patient. The result at 4 years is shown in Fig. 25.1C.

Splinting also can be used when the mandibular incisors are mobile because of severe bone loss. The same general steps are followed as described earlier. If further reinforcement is deemed necessary, however, *a plasma-coated woven polyethylene strip* such as **Ribbond** (Ribbond Inc., Seattle, WA) can be used to strengthen the splint. Additionally, the use of flowable composites greatly facilitates the placement of interproximal composite connectors. Fig. 25.2 illustrates a typical case.

- Following isolation with a rubber dam, small spaces (approximately 0.5 mm in width) are created between teeth with a flame-shaped diamond instrument to enable cross-sectionally strong composite connectors (Fig. 25.2A–C).
- Because a fibre-reinforcing material will be used, the *lingual surfaces to be bonded also should be lightly roughened* with an oval diamond to enhance the resin bonds.
- All interproximal and lingual surfaces to be bonded are etched for 15 s with a phosphoric acid-etching gel (Fig. 25.2D), followed by thorough rinsing and drying.
- Round wooden wedges can be used to stabilize the mobile teeth and to help maintain an open gingival embrasure form.
- To prevent any resin from sticking to the wooden wedges, a light coat of petroleum jelly can be placed on the wedges prior to positioning the wedges interproximally. Bonding agent is applied and cured to all etched surfaces (Fig. 25.2E).
- The interproximal composite connectors are then generated by injecting flowable composite into these areas and shaped (if needed) with a No. 2 explorer (Fig. 25.2F). A small amount of flowable composite is placed onto the lingual surfaces (but not cured) to receive the auxiliary splinting strip.
- An appropriate length of splinting material (polyethylene-coated woven fabric) is cut and first saturated with bonding agent. Then, by using a gloved finger, the strip is pressed into uncured composite and cured initially into place (Fig. 25.2G).
- The bonded strip is then covered incrementally with flowable composite, resulting in a smooth lingual surface (Fig. 25.2H). Facial and incisal embrasures are defined with the finishing burs to enhance esthetics.
- After finishing procedures, the rubber dam is removed and the occlusion is evaluated. The final result is seen in Fig. 25.2I and J.

Stabilization of Teeth After Orthodontic Treatment

After orthodontic treatment, teeth may require stabilization with either fixed or removable appliances. The latter method allows continued minor movements for the final positioning of teeth. When this position is reached, it is better to stabilize teeth with a fixed retainer. Removable retainers tend to irritate soft tissue. Also, they may be damaged, lost or not worn, which usually leads to undesired movement of teeth.

Fig. 25.3A shows a patient with a removable orthodontic retainer. Optimal positioning of teeth has been achieved by orthodontic movement; however, stabilization of teeth is required, and the unattractive spaces caused by undersized maxillary teeth need to be closed (Fig. 25.3B). A carefully planned appointment is required to accomplish the following:

I. remove any fixed orthodontic appliance,
II. add composite to close the diastemas and
III. stabilize teeth with a twisted stainless steel wire and composite.

Technique

- After the orthodontic appliance is removed and routine procedures are followed for closing the diastemas (Fig. 25.3C), the occlusion is examined carefully to determine the best position for locating the twisted wire because it will be placed only on the lingual surfaces.
- A sufficient length of twisted stainless steel wire (i.e. 0.45 mm [0.0175 in.] in diameter) is adapted to the

lingual surface of anterior teeth. A stone cast is helpful for adapting the wire.

- The wire must rest against the lingual surfaces passively without tension or interference with the occlusion. In the mouth, waxed dental tape is used to position the wire against teeth and hold it in place while the occlusal excursions are evaluated. The wire is attached only to the lingual fossa of each tooth.
- After the position of the wire has been determined, it is removed and only the enamel in the fossae (not the marginal ridges or embrasures) is etched, rinsed and dried.



Figure 25.2 Splinting of mobile mandibular incisors reinforced with a plasma-coated, polyethylene-woven strip (Ribbond; Ribbond Inc. Seattle, WA). (A and B) Facial and lingual preoperative views of mobile mandibular incisors that need splinting. (C) Preparation consists of roughening proximal surfaces and creating slight interdental spaces to provide bulk to the connector areas of the composite splint. (D) All interproximal and lingual surfaces to be bonded are etched with a phosphoric acid gel. (E) Teeth are stabilized with wooden wedges, and a bonding agent is applied. (F) Interproximal composite connectors are generated by injecting flowable composite. (G) A fibre-reinforcing strip is pressed into the uncured composite on lingual with a gloved finger. (H) The bonded strip is covered incrementally with flowable composite. (I and J) Completed fibre-reinforced composite-bonded periodontal splint seen from facial and lingual views.

(*Continued*)



Figure 25.2 (*cont.*)

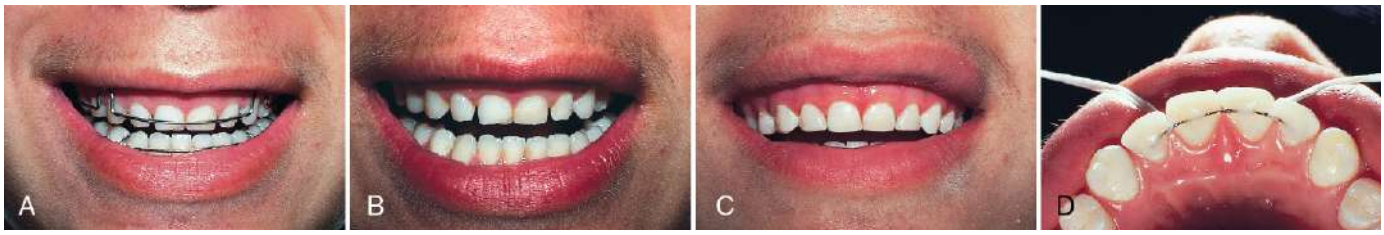


Figure 25.3 Stabilizing teeth after orthodontic treatment. (A) Patient with existing removable retainer. (B) Residual spaces resulting from undersized teeth. (C) Closure of spaces with composite additions is completed. (D) Orthodontic wire is held in position with dental tape and bonded into place with composite.

- Light-cured composite is best used for attaching the fixed wire splint. The wire is repositioned and held in place with a dental tape, while a sparing amount of resin-bonding agent is applied and lightly blown with air.
- After polymerization of the adhesive, a small amount of composite material is placed to encompass the wire in each fossa and bond it to the enamel. The operator must be careful not to involve the proximal surfaces (Fig. 25.3D). After polymerization of composite, the occlusion is evaluated and adjusted, as needed, for proper centric contacts and functional movements.

This unique splint allows some physiologic movement of teeth, yet it holds them in the correct position. The splint should remain in place for *at least 6 months* to ensure stabilization. Longer retention may be necessary, depending on the individual situation and recommendations of the orthodontist.

Avulsed or Partially Avulsed Teeth

Facial injuries often involve the hard and soft tissues of the mouth. The damage may range from lacerations of soft tissue to fractures of teeth and alveolar bone. Partial or complete avulsion of teeth can occur. Maxillary central incisors are involved more often, than are other teeth. A thorough clinical examination of soft tissue, lips, tongue and cheeks should be made to locate lacerations and embedded tooth fragments and debris. Radiographic examination is necessary to diagnose deeply embedded fragments or root fractures.

Treatment of soft tissue lacerations should include lavage, conservative debridement and suturing. Consultation with or referral to an oral surgeon may be necessary. A partially avulsed tooth is repositioned digitally and may or may not need splinting.

Traumatically avulsed teeth that are reimplanted immediately or within 30 min have a good prognosis for being retained.[1,2] After 30 min, the success rate declines rapidly. The avulsed tooth should be repositioned as soon as possible. In the interim, it should be placed in a moist environment such as saliva (i.e. held in the cheek or under the tongue), tooth-saver solution (i.e. Hank' s balanced solution), milk, saline or a wet towel. The replacement of avulsed teeth has immediate psychologic value and maintains the natural space in the event that a fixed prosthesis is required later.

Technique

- The maxillary right incisors that were completely avulsed in an accident (Fig. 25.4A) are repositioned immediately. After the teeth are repositioned, radiographs reveal that no other complications exist. Isolation with cotton rolls or gauze is preferable to the use of a rubber dam, which could cause malpositioning of the loose teeth. The occlusion should be evaluated to ensure that the teeth are properly positioned.
- The facial surfaces of the crowns are quickly cleaned with hydrogen peroxide, rinsed and dried by blotting with a gauze or cotton roll or by lightly blowing with air. The dentist should avoid blowing air into areas of avulsion or deep wounds to prevent air emboli.
- If a crown is fractured, deeply exposed dentin may need to be protected with a liner or base material.
- A twisted orthodontic wire (0.49 mm [0.0195 in.]) must be long enough to cover the facial (or lingual) surfaces of enough teeth to stabilize the loose teeth. The wire is adapted and the ends rounded to prevent irritation to soft tissue. In an emergency, a disinfected paper clip can be used as a temporary splint.
- No preparation of the enamel surface is necessary other than that provided by acid-etching. The middle third of the facial surfaces are etched, rinsed and dried of all visible moisture.
- Drying should be accomplished by blotting with a gauze or cotton roll and a light stream of air. Self-cured or light-cured composite may be used. The wire is positioned and held lightly in place, and the ends are attached with composite material (Fig. 25.4B).
- Light pressure is applied to the repositioned teeth as the facial surfaces are bonded to the wire in succession. Care is exercised not to allow composite to flow into the proximal areas. When the teeth are stabilized, any fractured areas can be conservatively repaired by the acid-etch and resin-bond technique.
- Finishing is accomplished by a flame-shaped carbide finishing bur and abrasive disks. The occlusion is evaluated carefully to ensure that no premature contacts exist.
- The patient is advised to maintain gentle care of the involved teeth. Antibiotic therapy may be required if the alveolar bone is fractured or a significant soft tissue damage has occurred. Tetanus shots or boosters are advised, if indicated by the nature of the accident; the patient's physician should be contacted about this.
- Appointments are made for *follow-up examinations on a weekly basis* for the first month.
- The patient is warned about symptoms of pulpal necrosis and advised to call if a problem develops. If root canal therapy is required, it is better accomplished with the splint in position.
- Removal of the splint is accomplished in *4–8 weeks* provided that recall visits have shown normal pulp test results and the teeth are asymptomatic. The wire is sectioned, and the resin material is removed with a flame-shaped, carbide finishing bur at high speed with air-water spray and a light intermittent application. Abrasive disks are used to polish the teeth to a high lustre.

Conservative Bridges

In selected cases, conservative bridges can be made by bonding a pontic to the adjacent natural teeth. These conservative bridges are classified according to the type of pontic:

I. natural tooth pontic,
II. denture tooth pontic,

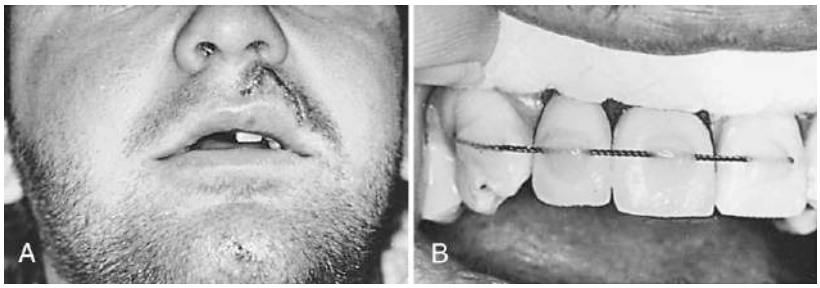


Figure 25.4 Splinting avulsed teeth. (A) Patient with traumatically avulsed maxillary right incisors. (B) Completed splint stabilizes repositioned incisors.

III. porcelain-fused-to-metal pontic or all-metal pontic with metal retainers and
IV. all-porcelain pontic.

Although the four types differ in the degree of permanency, they share a major advantage—conservation of the natural tooth structure. In addition, they can be viable alternatives to the conventional fixed bridges in circumstances where age, expense and clinical impracticality are considerations.

Because of the conservative preparation and bonded nature of all of these bridge types, retention is never as strong as in the case of a conventional bridge. As part of informed consent, patients should be told of the risk, although remote, of swallowing or aspirating bonded bridges that are dislodged. To reduce the risk of dislodgment, patients should be cautioned not to bite hard foods or objects with bonded bridge pontics.

The ideal site for a conservative bridge is where the edentulous space is no wider than one or two teeth. Other considerations include bite relation, oral hygiene, periodontal condition and extent of caries, defects and restorations in the abutment teeth. Conservative bridges are especially indicated for young patients because the teeth usually have large pulp chambers and short clinical crowns. Many older patients with gingival recession and mobile teeth are prime candidates because splinting can be incorporated with the bridge. More specific indications and clinical procedures for each of the four types of bridges are presented in the following sections.

I. Natural Tooth Pontic

The crowns of natural teeth (primarily incisors) often can be used as acid-etched, resin-bonded pontics. Considerations for this type of treatment include the following:

- periodontally involved teeth warrant extraction,
- teeth have fractured roots,
- teeth are unsuccessfully reimplanted after avulsion and
- root canal treatment has been unsuccessful.

However lost, the immediate replacement of a natural anterior tooth has a great psychologic value for most patients, although the procedure may be temporary. Natural tooth pontics also can be placed as interim restorations until an extraction site heals if conditions require a conventional bridge or an implant.

Certain prerequisites must exist to ensure a successful result:

I. the extracted tooth and abutments must be in reasonably good condition, especially the pontic, because it may become brittle and more susceptible to fracture;
II. the abutment teeth should be fairly stable and
III. the pontic must not participate in heavy centric or functional occlusion.

Because of this third restriction, canines and posterior teeth are not usually good candidates for this procedure. If the adjacent teeth are mobile, it is frequently necessary to secure them by splinting with composite (see the section on Techniques for Splinting Anterior Teeth).

Technique

- A maxillary right central incisor is being extracted for periodontal reasons (Fig. 25.5A and B). Before the tooth is extracted, a small round bur is used to place a shallow identifying mark on the facial surface to indicate the level of the gingival crest.
- If the tooth to be extracted is well-positioned in the dental arch, a PVS (Polyvinyl Siloxane) bite registration or putty impression material can be used to generate an index to reposition the natural tooth pontic in the correct pre-extraction position. After extraction, a 5 cm × 5 cm (2 in. × 2 in.) sponge is held in the space with pressure for haemorrhage control.
- By using a separating disk or a diamond instrument, the extracted tooth is transversely cut a few millimetres apical to the identification mark. When pontic length is determined, shrinkage of the healing tissue underlying the pontic tip must be anticipated. The root end is discarded.
- If the pulp canal and chamber have completely calcified, the next procedure is shaping and polishing the apical end of the natural tooth pontic as described in the following paragraphs. If the chamber is calcified as disclosed on the radiograph and the canal is nearly calcified, the canal is opened from the apical end by using a small round bur or diamond to the extent of the canal.
- The operator should be as conservative of the tooth structure as possible and yet provide access for subsequent injection of the composite material to fill the canal. A large chamber and canal are instrumented and debrided using conventional endodontic procedures with access from the apical end (Fig. 25.5C).
- Access is provided for subsequent injection of composite. Removal of the pulpal tissue in this manner prevents discolouration of the tooth caused by degeneration products. Traditional lingual access for instrumentation is avoided to prevent weakening the pontic. After these procedures, the canal (and chamber, if present) is filled and closed with composite.
- After composite has been fully polymerized, the apical end is contoured to produce a bullet-shaped ovate design (Fig. 25.5C). This design provides adaptation of the pontic tip to the residual ridge, and yet it allows the tissue side of the pontic tip to be cleaned with dental floss. It is also the *most esthetic pontic tip design* that can be used.
- While being contoured, the tip is occasionally evaluated by trying the pontic in the space. In the maxillary



Figure 25.5 Resin-bonded maxillary natural tooth pontic. (A) Preoperative photograph before extraction of periodontally involved maxillary right central incisor. (B) Extraction site immediately after the removal of an incisor. (C) Enlarged apical opening ready to be filled with composite. The pontic tip has been contoured to an ovate design. (D) The abutment teeth are isolated, roughened and acid etched. (E) Immediate postoperative photograph of natural tooth pontic bonded in place. (F) Resin-bonded natural tooth pontic with healed residual ridge 6 weeks later.

arch, passive contact between the pontic tip and the healed residual ridge is considered ideal for maximal phonetic and esthetic potential. In the mandibular arch (where esthetics is not generally a problem), the pontic tip is best shaped into the same bullet-shaped design but positioned as a hygienic pontic type that does not contact tissue (Fig. 25.6A). The pontic tip is smoothed and polished using a proper sequence of abrasive disks or polishing points. A polished pontic tip not only is easier to clean but also retains less plaque.

- Usually, a rubber dam is needed for isolation of the region to prevent seepage of blood and saliva. Isolation using cotton rolls and gingival retraction cords is acceptable if the haemorrhage has been controlled. Any caries lesions or faulty proximal restorations on involved proximal surfaces of the pontic and the abutments are restored with light-cured composite (preferably the same material to be used subsequently for the bridge connectors) by using modified preparation designs. It is recommended that the resulting restored surfaces be under-contoured rather than over-contoured to facilitate positioning of the natural tooth pontic.
- Next, the involved proximal surfaces on the abutment teeth and the pontic are roughened with a coarse, flame-shaped diamond instrument. Spaces of approximately 0.5 mm should exist between the pontic and the abutment teeth, because stronger connectors are provided by the additional bulk of the composite material. Now, the operator should acid-etch, rinse and dry all the prepared (i.e. roughened) surfaces (Fig. 25.5D).
- Light-cured composite is preferred for bonding natural tooth pontics because the extended working time allows the operator to contour the connectors before polymerization. First, the adhesive is applied to the etched surfaces of the pontic and lightly blown with air to remove the excess. Then it is polymerized by application of light, and the pontic is set aside (ready for bonding in the mouth). Next, the adhesive is applied to the etched surfaces of the abutment teeth and cured. A small amount of composite material is placed on the proximal contact areas of the natural tooth pontic, and the pontic is inserted carefully in the proper position in the mouth. The composite is shaped around the

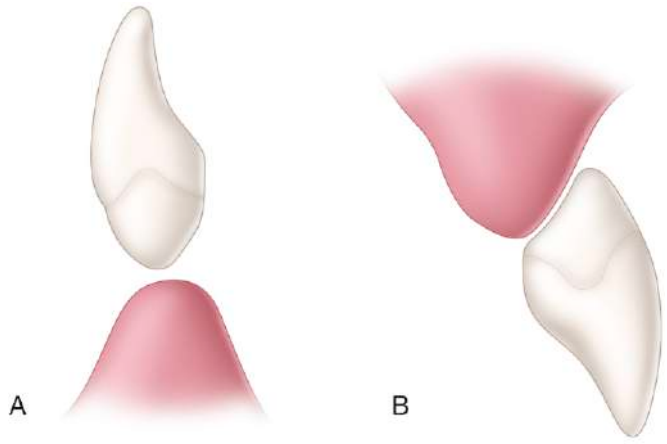


Figure 25.6 Pontic tip design. (A) Hygienic-type pontic with ovate or bullet-shaped tip. (B) Modified ridge lap-type pontic with slight concavity conforming to residual ridge.

contact areas with an explorer tip. After final verification that the pontic position is correct, composite is polymerized with light. Next, additional composite is applied in the proximal areas (more material is added on the lingual than on the facial surface), contoured and cured. Adequate gingival embrasures must be provided to facilitate flossing and ensure gingival health. After sufficient material has been added and polymerized, the embrasure areas should be shaped and smoothed with carbide finishing burs or fine diamonds and polishing disks or points.

- The rubber dam is removed, and the occlusion is evaluated for centric contacts and functional movements. Heavy contacts on the pontic or the connector areas must be adjusted. The finished bridge immediately after bonding is illustrated in Fig. 25.5E.
- The patient should return in 4–6 weeks for evaluation of the relationship of the pontic tip to the tissue. Passive contact should exist between the pontic tip and the underlying tissue to prevent ulceration. If tissue ulceration is present, the pontic must be removed, recontoured and rebonded. The finished bridge and healed residual ridge are shown in Fig. 25.5F.

As stated earlier, abutment teeth that are mobile often can be splinted with composite to afford stability to periodontally involved teeth. The abutments are isolated, roughened and acid-etched (Fig. 25.7A). Because esthetics is not as crucial, a hygienic pontic tip is recommended for mandibular incisors (Fig. 25.6A). The finished bridge splint is illustrated in Fig. 25.7B.

II. Denture Tooth Pontic

An acrylic resin denture tooth can be used as a pontic for the replacement of missing maxillary or mandibular incisors by using the resin-bonding technique (Fig. 25.8A–H).

Although this type of bridge is sometimes used as an interim prosthesis and is called a *temporary bridge*, it can be a viable alternative to a conventional bridge and may last for years in some circumstances.

As with the natural tooth pontic, the *major contraindications* to this type of resin-bonded bridge are:

I. abutment teeth that have extensive caries or restorations,
II. abutment teeth that have mobility and
III. a pontic area that is subjected to heavy occlusal forces.

In the illustrated example, the permanent maxillary right lateral incisor is missing and the adjacent teeth are in favourable condition and position (Fig. 25.8A). Further examination reveals an ideal situation for a conservative bridge that uses a denture tooth pontic.

Technique

- Although the entire procedure can be completed at chairside in one appointment, considerable time can be saved by an indirect technique. During the first appointment, the shade (Fig. 25.8) and mold of the denture tooth are selected, and impressions are made.
- In the laboratory, stone casts are poured, and the ridge area is relieved slightly and marked with a soft lead pencil. As the pontic is trial positioned, the pencil markings rub off onto its tip to facilitate contouring of this area (Fig. 25.8C).
- Contouring is best accomplished with acrylic burs and a Burlew wheel in a straight handpiece. The tissue side of the pontic should be contoured to a modified ridge lap configuration that is convex mesiodistally and slightly concave faciolingually (Fig. 25.6B).
- This type of design not only allows the pontic tip to adapt to the residual ridge, but it also allows for effective cleaning with dental floss. After it is contoured, the pontic tip should be smoothed and highly polished with pumice and an acrylic-polishing agent (Fig. 25.8D).
- Because composite does not normally bond to acrylic resin, provisions must be made to facilitate a strong connection between the pontic and the adjacent teeth. One provision may be completed in the laboratory by preparing large Class III conventional preparations in the pontic that mechanically retain the composite material. The outline of the preparations must be large

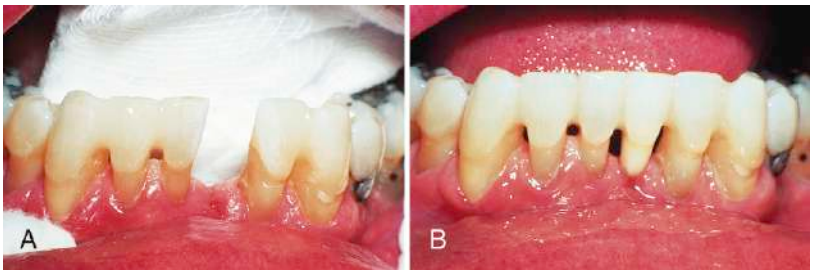


Figure 25.7 Resin-bonded mandibular bridge splint using natural tooth pontic. (A) The anterior segment is splinted with composite, and the abutment teeth are isolated, roughened and etched. (B) Natural tooth pontic is bonded in place.

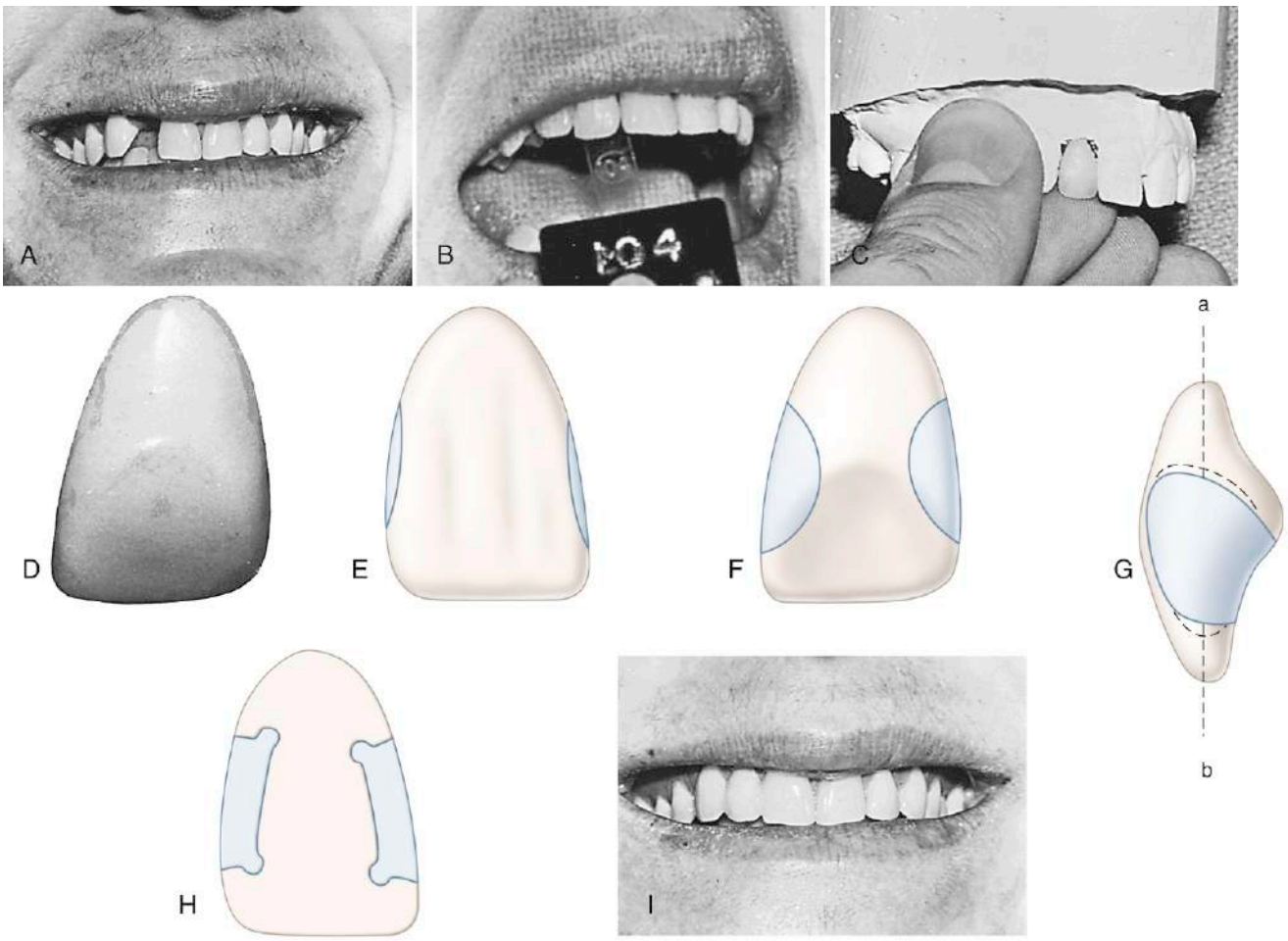


Figure 25.8 Resin-bonded denture tooth pontic. (A) Preoperative photograph shows a missing maxillary lateral incisor. (B) Shade and mold selection. (C) Positioning pontic on working model while contouring. (D) Contoured and polished pontic (lingual view). (E–G) Outline form of Class III preparations: facial (E), lingual (F) and proximal (G) views. (H) Cross section of denture tooth (longitudinal section) in plane *ab* as seen in (G) showing the mechanical retention form incisally and gingivally as prepared with a No. ½ bur. (I) Denture tooth pontic is bonded in place with composite.

enough to provide adequate surface area of the composite restoration for bonding to the adjacent teeth (Fig. 25.8E–G).

- An appropriately sized round bur (No. 2 or No. 4) is used to cut each preparation to a depth of approximately 1.5 mm and extend the outline approximately 0.5 mm past the contact areas into the gingival, incisal and facial embrasures. Even more extension should be made into the lingual embrasure to provide for bulk of composite material in the connector areas.
- The lingual extensions should not be connected because this unnecessary step would unduly weaken the pontic. Mechanical undercuts are placed at the incisoaxial and gingivoaxial line angles with a No. ½ bur to lock the composite material (to be inserted later in the technique) mechanically in the acrylic resin pontic (Fig. 25.8G and H).
- At the next appointment, the pontic is tried in place to confirm that the shade and contours are correct. Approximately 0.5 mm of space should exist between each proximal 'contact' and the abutment tooth. The pontic is cleaned with acetone to remove dust and debris.
- Retention of the pontic by undercuts, as previously described, also can be augmented by a second provision – the conditioning of the proximal aspects of the pontic with two applications of ethyl acetate, a polymer softener. A thin layer is applied in the Class III preparations and on the cavosurface areas and allowed to dry for 5 min. This process is repeated to ensure optimal bonding.
- The preparations are filled with the same light-cured composite material expected to be used for bonding the pontic in place. The composite should be applied and cured in the retentive areas before the remainder of the preparation is filled. This step ensures complete polymerization. After the entire preparation is filled, it should be polymerized again with the light source. It is better to leave the contact areas slightly under-contoured for the pontic to fit easily between the abutment

teeth. The pontic is set aside in a safe place for some time.

- Isolation of the abutment teeth should be accomplished with cotton rolls and retraction cords (rather than with a rubber dam) to relate the pontic better to the residual ridge area. Any caries or old restorations in the adjoining proximal areas of the abutment teeth should be removed at this time, and any indicated liners should be applied. The proximal surfaces of the abutment teeth are roughened with a coarse flame-shaped diamond instrument. This step is followed by acid-etching, rinsing and drying. The adhesive is applied, lightly blown with air and cured. Tooth preparations, if present, are restored with the same composite material. Care is taken not to over-contour the restoration or restorations.
- The pontic is evaluated by positioning it temporarily in the edentulous space. If adjustments are made, the surfaces should be cleaned with acetone. Next, a small amount of composite is wiped onto the contact areas (mesial and distal) of the pontic, and the pontic is placed into the proper position between the abutment teeth. An explorer tip is helpful in placing the material evenly around the contact area. Care must be taken to place the pontic so that it lightly touches the ridge, but does not cause tissue blanching. The composite material used to position the pontic is polymerized.
- It is helpful to add and cure the additional composite in small increments to obtain the correct contour and minimize finishing procedures. The facial, incisal and gingival embrasures should be defined with a flame-shaped finishing bur or fine diamond and polished with appropriate disks or points. The lingual aspect of the bridge is contoured with a round finishing bur without defining lingual embrasures because this could weaken the connectors. The retraction cords are removed from the gingival crevice. Articulating paper is used to mark the occlusion, and any offensive contacts are removed. The final restoration is shown in Fig. 25.8I.

III. Porcelain-fused-to-metal Pontic or All-metal Pontic with Metal Retainers

A stronger and more permanent type of acid-etched, resin-bonded bridge is possible by use of a cast metal framework.[3,4]

- In anterior areas where esthetics is a consideration, the design of the bridge includes a porcelain-fused-to-metal (PFM) pontic with metal winged retainers extending mesially and distally for attachment to the proximal and lingual surfaces of the abutment teeth.
- In posterior areas where esthetics is not a critical factor, the bridge can have either a PFM or an all-metal pontic.

The technique is more complicated and time consuming than the previously described methods because it requires some initial tooth preparation, an impression, laboratory procedures and a second appointment for etching and bonding. Compared with conventional bridges, resin-bonded bridges of this type offer five distinct advantages:

I. Anaesthesia is usually not required.
II. The tooth structure is conserved (i.e. no dentin involvement).
III. Gingival tissues are not irritated because margins usually are not placed sub-gingivally.
IV. An esthetically pleasing result can be obtained more easily.
V. The cost is lower because less chair time is required, and laboratory fees are lower as well.

Ideally, this type of conservative bridge is used for short spans in the anterior or posterior areas with sound abutment teeth in good alignment. The most favourable occlusal relationship exists where little or no centric contact and only light functional contact are present. However, teeth can be prepared and the bridge framework designed to withstand moderately heavy occlusal forces. Orthodontics may be required to improve tooth alignment. The bridge also can be extended to splint adjacent periodontally involved teeth. Surgical crown-lengthening procedures sometimes are indicated for teeth with short clinical crowns.

Although minimal, some preparation of the enamel of the abutment teeth is mandatory in the retainer area of the bridge to:

- provide a definite path of insertion or seating or both,
- enhance retention and resistance forms,
- allow for the thickness of the metal retainers and
- provide physiologic contour to the final restoration.

The importance of the tooth preparation design cannot be overemphasized. The success of these types of bridges depends on the preparation design. The bridges must be independently retentive by design and cannot rely solely on resin bonding for retention. Preparation design for these types of bridges is similar to that for a cast three quarter crown; however, it is restricted to enamel.

The preparation for each abutment varies, depending on the individual tooth position and anatomy. Approximately the same amount of surface area should be covered on each abutment tooth. In some situations, recontouring of the adjacent and opposing teeth may be indicated. The details of the preparations are described later.

Two primary types of resin-bonded bridges with metal retainers currently exist[3,4]:

i. Rochette Resin-bonded Bridges with Metal Retainers

The Rochette type uses small countersunk perforations in the retainer sections for retention and is best suited for anterior bridges (Fig. 25.9A).[4] Care must be exercised in placing the perforations to prevent weakening the framework. Perforations that are too large or too closely spaced invite failure of the metal retainer by fracture. The perforations

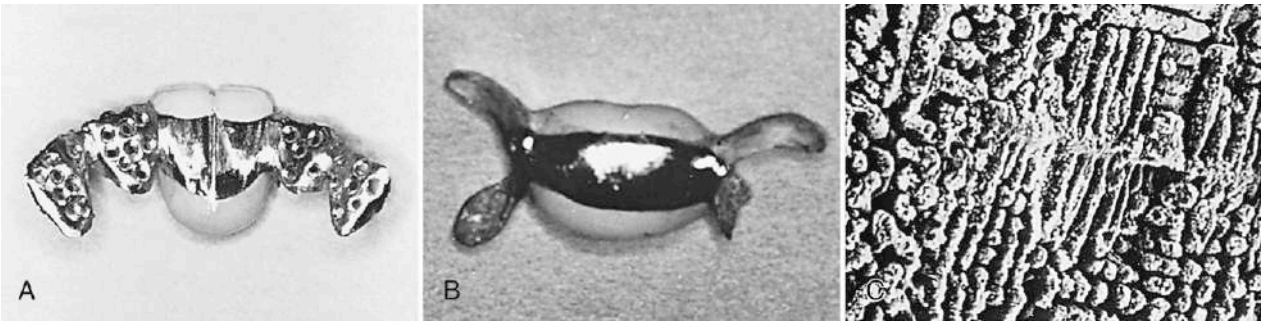


Figure 25.9 Acid-etched, resin-bonded metal bridges. (A) Rochette type. (B) Maryland type. (C) Scanning electron micrograph of etched metal surface.
(Courtesy: Dr John Sturdevant.)

should be approximately 1.5 to 2 mm apart and have a maximum diameter of 1.5 mm on the tooth side. Each hole is countersunk so that the widest diameter is towards the outside of the retainer. When the bridge is bonded with a resin cement, it is mechanically locked in place by microscopic undercuts in the etched enamel and the countersunk holes in the retainer (Fig. 25.10A).

The advantages of this design include the following:

- It is easy to see the retentive perforations in the metal.
- If the bridge must be removed or replaced, the bonding medium can be cut away in the perforations to facilitate easy removal.
- No metal etching is required.

The disadvantages of this design include the following:

- The perforations, if improperly sized or spaced, could weaken the retainers.
- The exposed resin cement is subject to wear.
- It is not possible to place perforations in proximal or rest areas.

ii. Maryland Resin-bonded Bridges with Metal Retainers

A second type of cast metal framework, commonly known as the Maryland bridge, is reported to have improved bonding strength (Fig. 25.9B).[3,5] Instead of perforations, the *tooth side of the metal framework is electrolytically or chemically etched*, which produces microscopic undercuts (Fig. 25.9C).

The bridge is attached with a self-cured, resin-bonding medium that locks into the microscopic undercuts of the etched retainer and the etched enamel (Fig. 25.10B). It can be used for anterior and posterior bridges. Although this design has been reported to be stronger, it is more technique sensitive because the retainers may not be properly etched or may be contaminated before cementation. Because the retentive features cannot be seen with the unaided eye, the etched metal surfaces must be examined under a microscope to verify proper etching (minimum magnification).

More recently, Maryland bridges have been fabricated with no electrolytic etching of the surface and chemically bonded to the tooth after a process called *silicoating* or with a 4-META or phosphate ester-containing, resin-bonding medium.[6,7] Resin materials containing 4-META or other resin monomers are capable of strongly bonding to metal surfaces.[8,9] Surface roughening with micro-etching (i.e. sandblasting) is commonly used in conjunction with these adhesive cements. These types of Maryland bridges are referred to as *adhesion bridges* and differ only in the means of retention. The design of adhesion bridges is the same for this alternative Maryland bridge design.

Successes and failures have been observed with both bonded bridge designs. Because the procedures are technique-sensitive, every step must be followed carefully.

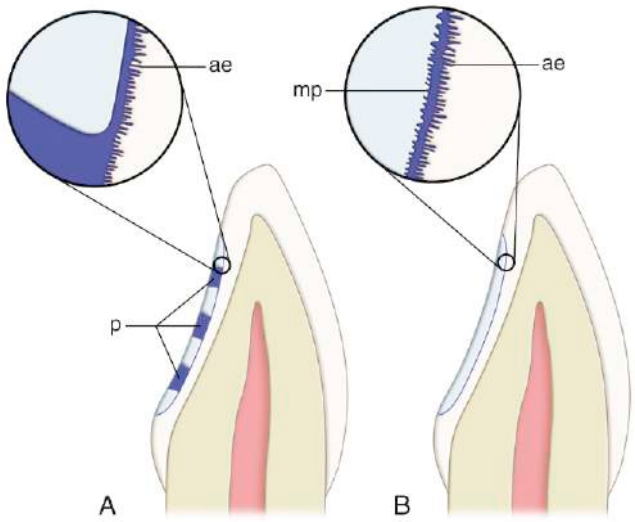


Figure 25.10 Cross-sectional diagram of two types of resin-bonded bridges. (A) In addition to acid-etching prepared enamel surfaces (*ae*), the Rochette type uses small countersunk perforations (*p*) in the retainer section. (B) In the Maryland type, the tooth side of the framework is either etched to produce microscopic pores (*mp*) or bonded with no etching with an adhesive cement.

IV. All-porcelain Pontic

Improvements in dental porcelains along with the capacity to etch and bond strongly to porcelain surfaces have made all-porcelain pontics a viable alternative to pontics with metal winged retainers (e.g. Maryland and Rochette bridges).[10,11]

- Although all-porcelain pontics are not as strong as pontics with metal retainers, far superior esthetic results can be achieved because no metal substructure or framework is present.
- All-porcelain pontics often can be used when tooth anatomy precludes or restricts the preparation and placement of a metal winged pontic. Long, pointed canines with proximal surfaces exhibiting little occlusogingival height often lack adequate areas for the placement of retention grooves.
- Anterior teeth that are notably thin faciolingually also are not good candidates for metal, resin-bonded bridge retainers and often are esthetic failures because of metal showing through the tooth. In such instances, custom-fabricated, etched porcelain pontics frequently can provide an esthetic, functional alternative.
- All-porcelain pontics are particularly indicated in adolescents and young adults, in whom virgin, unrestored teeth are often encountered. Because teeth are not extensively prepared, this procedure is almost entirely reversible. This is a major benefit in young patients, where all-porcelain pontics can be placed as interim restorations until implants or a more permanent prosthesis can be placed at an older age.

Because of their limited strength, all-porcelain pontics should be considered provisional in nature, similar to the natural tooth pontic and the acrylic denture tooth pontic. Similar to the natural tooth and denture tooth pontics, certain prerequisites must be met to ensure a successful result:

I. First, the abutment teeth must be in reasonably good condition with proximal enamel surfaces that are intact or contain very small composite restorations.
II. Second, the abutment teeth should be stable with little mobility present. If the abutment teeth are mobile, it is frequently necessary to secure them as well by splinting with composite to adjacent teeth before placement of the bonded pontic (see section Techniques for Splinting Anterior Teeth).
III. Third, the pontic must not be placed in a position that would subject it to heavy centric or functional occlusal contacts. Because of these occlusion concerns, canines and posterior teeth are not usually good candidates for these types of resin-bonded bridges.

Technique

Fig. 25.11A and B illustrates a typical case of congenitally missing lateral incisors in which tooth contours contraindicated the use of resin-retained bridges with metal retainers. Central incisors are very translucent, and the mesial contours of canines are deficient (Fig. 25.11C and E). After assessing centric and functional occlusions, it was determined that all-porcelain pontics could be placed without subjecting them to heavy occlusal forces.

- At the first appointment, the involved abutments are cleaned with flour of pumice and an accurate shade selection is made, noting any desired colour gradients or characterizations.
- No preparation of the teeth is recommended unless the proximal surfaces of the abutment teeth adjacent to the edentulous space are markedly convex. In such cases, slight flattening of the proximal surfaces with a diamond instrument facilitates closer adaptation of the pontic to the abutment teeth, increasing strength of the connectors. Otherwise, no retentive features are recommended for the preparation in the abutment teeth; the connector areas are entirely made of composite.
- Bridge connectors composed of porcelain are subject to eventual fatigue fracture, after which repair is made more difficult. Studies show that *'veneer bridges'* (i.e. all-porcelain pontics retained by adjacent etched porcelain veneers) in particular are the weakest design of all and should be avoided.[11] These types of bridges not only provide little bond strength to the pontic but also needlessly cover adjacent, healthy facial tooth surfaces. All-porcelain pontics (composite used for bonding to the abutment teeth) are similar to extracted natural tooth pontics in this regard. Those that have connector areas consisting of the design feature allow for easy repair and replacement of the composite connector should a fracture in this area be encountered.
- If high-strength ceramics that are totally immune to crack propagation and cohesive fracture are developed, retentive features prepared in the adjacent abutment teeth may be desired. These features, prepared in enamel, would consist of *proximal grooves* or *boxes*, depending on the faciolingual dimension of the proximal surfaces. In the absence of such totally fracture-resistant ceramics, however, all-porcelain pontics are best placed with composite connectors for ease of repair and replacement.
- An elastomeric impression is made, and a working cast is generated from it. A modified ridge lap pontic tip design (Fig. 25.6B) is recommended. An occlusal bite registration should be made and forwarded to the laboratory so that the occlusal relationship can be considered during fabrication of all-porcelain pontics.
- The proximal surfaces of the pontics are etched with hydrofluoric acid. The area etched must include all areas anticipated for bonding to the composite-bonding medium. The etched proximal surfaces should extend just beyond the lingual line angles so that additional



Figure 25.11 All-porcelain pontics. (A and B) Patient with congenitally missing lateral incisors. (C and D) Right side before and after treatment. (E and F) Left side before and after placement of all-porcelain pontic. (G) Lingual view of completed bridges. (H) Facial view of all-porcelain pontics.

composite can be placed in the lingual embrasure areas for additional connector strength.

- At the subsequent appointment, teeth are isolated with cotton rolls. A 5 cm × 5 cm (2 in. × 2 in.) cotton gauze is placed across the back of the patient's mouth to act as a protective shield should the pontic be inadvertently dropped. A rubber dam is not recommended for this procedure because it precludes accurate assessment of the adaptation of the pontic tip to the residual ridge.
- Before the teeth dehydrate, the position of each pontic is tested in the edentulous space to assess the shade and relationship of the pontic tip to the residual ridge. The pontic tip should contact the residual ridge passively with no blanching of the underlying tissue evident. Spaces of approximately 0.3–0.5 mm should exist between the pontic and the abutment teeth because stronger connectors are provided by the additional bulk of composite material.
- Care must be taken not to allow contamination of the etched pontic from saliva to occur during the try-in phase. If saliva contamination occurs, the etched proximal surfaces of the pontic must be cleaned thoroughly with alcohol and dried. After try-in, all etched proximal surfaces of the porcelain pontics are primed with a suitable silane-coupling agent (see the manufacturer's instructions for the specific technique). The pontics are now ready for bonding.
- The involved proximal enamel surfaces of the abutment teeth are roughened with a coarse, flame-shaped diamond instrument. Thereafter, all of the prepared (i.e. roughened) enamel surfaces should be acid etched, rinsed and dried. Care must be taken to maintain clean, dry, uncontaminated etched surfaces until the pontic is positioned and bonded. The abutment teeth are now ready for bonding.
- A light-cured composite is preferred for bonding all-porcelain pontics because the extended working time allows the operator to contour the connectors initially before polymerization. The dentist applies the adhesive to the etched surfaces of the porcelain pontic and the abutment teeth and lightly blows with air to remove the excess. A 20-s application of light from the light-curing unit is used to polymerize the bonding agent on each etched surface.
- A small amount of composite material is placed on the proximal contact areas of the natural tooth pontic and the pontic is inserted carefully into the proper position in the edentulous space. A stent, or index, made from bite registration material or fast-setting plaster can be used to position the pontic, if desired. Positioning by hand is recommended, however, so that optimal gingival pressure can be maintained for best tissue adaptation.
- The dentist shapes the excess composite extruding from the connector areas around the contact areas with an explorer tip or small plugger end of a composite instrument. After final verification that the pontic position is correct, the composite is polymerized with light for a minimum of 40–60 s each from facial and lingual directions (for a total of 80–120 s).
- Additional composite is applied in the proximal areas (more material is added on the lingual surface than on the facial surface), contoured and polymerized. Adequate gingival embrasures must be maintained to facilitate flossing and ensure good gingival health. After sufficient material has been added and polymerized, the dentist shapes and smoothes the embrasure areas with carbide finishing burs, fine diamonds and polishing disks. Facial embrasures are defined for esthetics, but lingual embrasures are closed with composite to strengthen the connectors (Fig. 25.11D, F and G).
- The dentist evaluates the occlusion centric contacts and functional movements. Heavy contacts on the pontic or the connector areas must be adjusted. The finished bridges (immediately after bonding) are illustrated in Fig. 25.11D and F–H. As with all resin-bonded bridges, patients must be advised to avoid biting into hard foods or objects to reduce the risk for dislodgment.

Also, the patient must be advised as part of informed consent that although the chances are remote, the potential for dislodgment and the risk of swallowing or aspirating the pontic do exist. This possibility exists for all resin-bonded bridges, and patients must be warned of this hazard even though the risk is minimal.

Summary

Stabilization of mobile teeth due to trauma, periodontal disease or after fixed orthodontic treatment using a resin-bonded splint is an important clinical procedure that a clinician should master. The use of a resin-bonded bridge should not be the primary treatment of choice for missing teeth; however, this offers a conservative and esthetic alternative in certain clinical situations.

References

1. Andreasen JO: The effect of pulp extirpation or root canal treatment on periodontal healing after replantation of permanent incisors in monkeys, *J Endod* 7:245, 1981.
2. O'Riorden MW, Ralstrom CS, Doerr SE: Treatment of avulsed permanent teeth: an update, *J Am Dent Assoc* 105:1028, 1982.
3. Livaditis G: Cast metal resin-bonded retainers for posterior tooth, *J Am Dent Assoc* 101:926, 1980.
4. Rochette AL: Attachment of a splint to enamel of lower anterior teeth, *J Prosthet Dent* 30:418, 1973.
5. Livaditis G, Thompson VP: Etched castings: an improved retentive mechanism for resin-bonded retainers, *J Prosthet Dent* 47:52, 1982.
6. Hamada T, Shigeto N, Yanagihara T: A decade of progress for the adhesive fixed partial denture, *J Prosthet Dent* 54:24, 1985.
7. Hansson O: The silicoater technique for resin-bonded prostheses: clinical and laboratory procedures, *Quintessence Int* 20:85, 1989.
8. Cooley RL, Burger KM, Chain MC: Evaluation of a 4-META adhesive cement, *J Esthet Dent* 3:7, 1991.
9. Matsumura H, Nakabayashi N: Adhesive 4-META/MMA-TBB opaque resin with poly(methyl methacrylate)-coated titanium dioxide, *J Dent Res* 67:29, 1988.
10. Heymann HO: The "Carolina Bridge": a novel interim all-porcelain bonded prosthesis, *J Esthet Restor Dent* 18(2): 81–91, 2006.
11. Moore DL, Demke R, Eick JD, et al: Retentive strength of anterior etched porcelain bridges attached with composite resin: An in vitro comparison of attachment techniques, *Quintessence Int* 20:629, 1989.

Page left intentionally blank

Chapter 26

Class III and V Amalgam Restorations

"Tell me and I forget. Teach me and I remember. Involve me and I learn"

—Benjamin Franklin

This chapter presents information about class III and V amalgam restorations. Class III restorations are indicated for defects located on the proximal surface of anterior teeth that do not affect the incisal edge. Part of the facial or the lingual surfaces may also be involved in class III restorations. Class V restorations are indicated to restore defects on the facial or lingual cervical one-third of any tooth.

The class III amalgam restoration is rarely used. Its use has been supplanted by tooth-coloured restorations (primarily composite), which have become increasingly wear-resistant and colour-stable. Because indications exist for class III amalgam restorations, however, practitioners should be familiar with this restorative technique.

The class V amalgam restoration can be especially technique-sensitive because of location, extent of caries, and limited access and visibility. Cervical caries usually develops because of the chronic presence of acidogenic plaque located in the non-self-cleansing area just beneath the coronal height of contour. Patients with gingival recession are predisposed to cervical caries because dentin is more susceptible to demineralization than enamel. Patients with a reduced salivary flow caused by certain medical conditions (e.g. Sjögren's syndrome), medications or head and neck radiation therapy are also predisposed to cervical caries. These patients usually have less saliva to buffer the acids produced by oral bacteria. Patients with gingival recession that has exposed the root surface have a predisposition to *root caries* because dentin is more susceptible to demineralization than enamel. Class V restorations may be used to treat both cervical and root caries lesions.

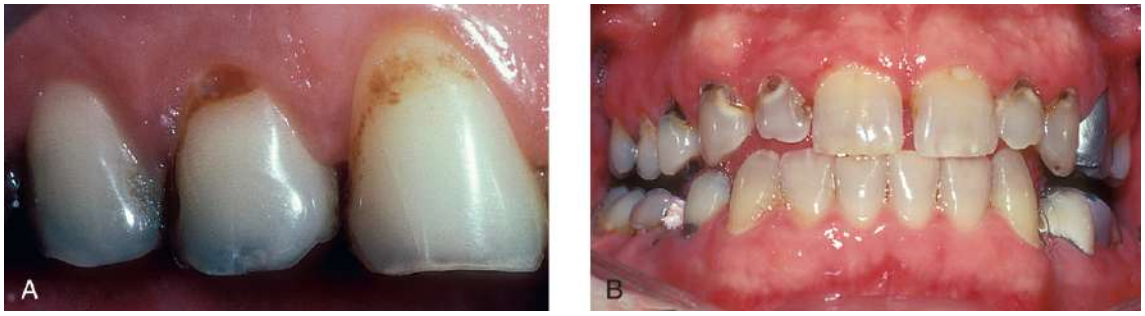
Incipient, smooth-surface enamel caries appears as a chalky white line just occlusal or incisal to the crest of the marginal gingiva (usually on the facial surface) (Online Fig. 26.1). These areas often are overlooked in the oral examination, unless teeth are free of debris, isolated with cotton rolls and dried gently with the air syringe. When incipient cervical caries has not decalcified the enamel sufficiently to result in cavitation (i.e. a break in the continuity of the surface), the lesion may be remineralized by appropriate techniques, including patient motivation towards proper diet and hygiene. Occasionally, an enamel surface that is only slightly cavitated may be treated successfully by smoothing with sandpaper disks, polishing and treating with a fluoride varnish or a dentin adhesive in an attempt to prevent further caries that may require treatment. This prophylactic, preventive treatment cannot be instituted if caries has progressed to decalcify and soften enamel to an appreciable depth. In this instance, a class V tooth preparation and restoration is indicated, particularly if caries has penetrated to the dentinoenamel junction (DEJ) (Online Fig. 26.2A). When numerous cervical lesions are present (Online Fig. 26.2B), a relatively high caries index is obvious. In addition to the restorative treatment, the patient should be instructed and encouraged to implement an aggressive prevention programme to avoid recurrent decay.



Online Figure 26.1 Incipient caries lesions of enamel appear as white spots. The affected surface may be smooth (i.e. non-cavitated). White spots are more visible when dried.
(From Cobourne MT, DiBiase AT: *Handbook of orthodontics*, Edinburgh, 2010, Mosby.)

Pertinent Material Qualities and Properties

Material qualities and properties important for class III and V amalgam restorations are strength, longevity, ease of use and past success. (See Chapter 12 for a discussion of the pertinent material qualities and properties of amalgam.)



Online Figure 26.2 Cervical caries. (A) Cavitation involving enamel and dentin in several teeth. (B) Relatively high caries index is obvious when numerous cervical lesions are present.
(From Perry DA, Beemsterboer PL: *Periodontology for the dental hygienist*, ed 3, St. Louis, MO, 2007, Saunders.)

Indications

Few indications exist for a class III amalgam restoration. It is generally reserved for the distal surface of maxillary and mandibular canines if (i) the preparation is extensive with only minimal facial involvement, (ii) the gingival margin primarily involves cementum or (iii) moisture control is difficult. For esthetic reasons, amalgam rarely is indicated for the proximal surfaces of incisors and the mesial surface of canines.

Class V amalgam restorations may be used anywhere in the mouth. As with class III amalgam restorations, they are generally reserved for non-esthetic areas, for areas where access and visibility are limited and where moisture control is difficult and for areas that are significantly deep gingivally. Because of limited access and visibility, many class V restorations are difficult and present special problems during the preparation and restorative procedures.

One measure of clinical success of cervical amalgam restorations is the length of time the restoration serves without failing (Online Fig. 26.3). Properly placed class V amalgams have the potential to be clinically acceptable for many years. Some cervical amalgam restorations show evidence of failure, however, even after a short period. Inattention to tooth preparation principles, improper manipulation of the restorative material and moisture contamination contribute to early failure. Extended service depends on the operator's care in following accepted treatment techniques and proper care by the patient.

Amalgam may be used on partial denture abutment teeth because amalgam resists wear as clasps move over the restoration. Contours prepared in the restoration to retentive areas for the clasp tips may be achieved relatively easily and maintained when an amalgam restoration is used. Occasionally, amalgam is preferred when the caries lesion extends gingivally enough that a mucoperiosteal flap must be reflected for adequate access and visibility (Online Fig. 26.4). Proper surgical procedures must be followed, including sterile technique, careful soft tissue management and complete debridement of the surgical and operative site before closure.

Contraindications

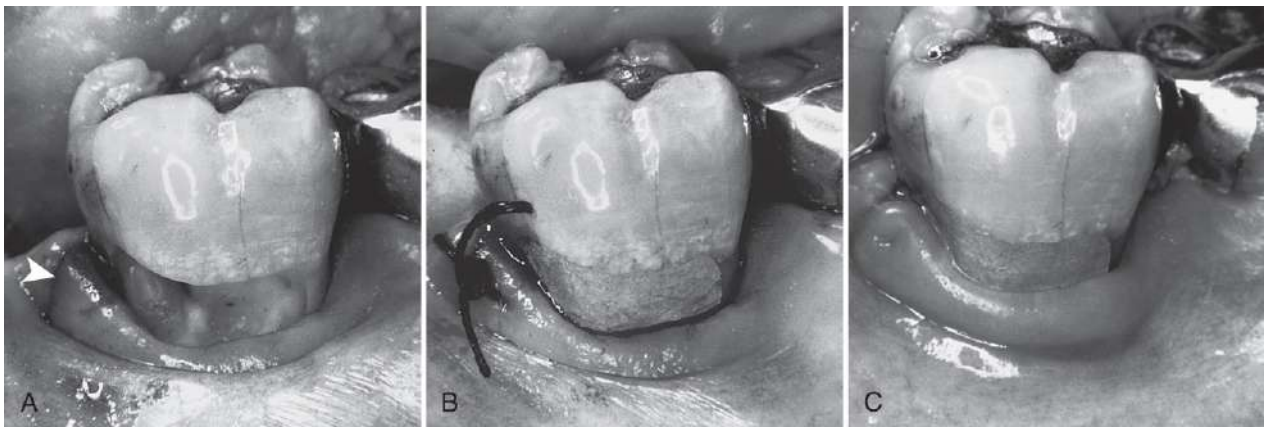
Class III and V amalgam restorations usually are contraindicated in esthetically important areas because many patients object to metal restorations that are visible (Online Fig. 26.5). Generally, class V amalgams placed on the facial surface of mandibular canines, premolars and molars are not readily visible. Amalgams placed on maxillary premolars and first molars may be visible. The patient's esthetic demands should be considered when planning treatment.

Advantages

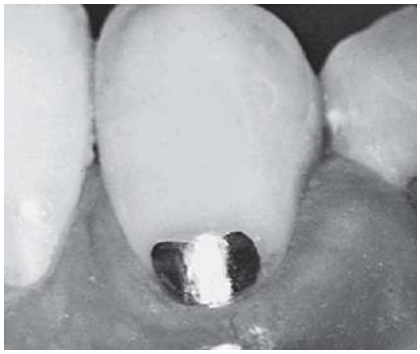
Amalgam restorations are stronger than other class III and V direct restorations. In addition, they are generally easier to place and may be less expensive to the patient. Because of its metallic colour, amalgam is easily distinguished from



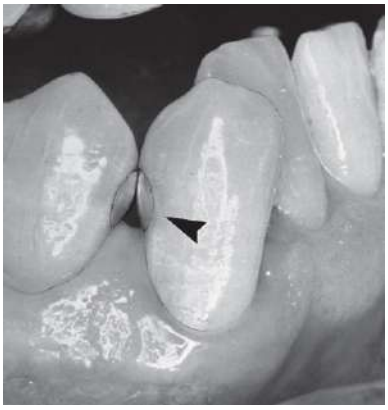
Online Figure 26.3 (A) Six-year-old cervical amalgam restorations. (B) After 16 years, some abrasion and erosion are evident at the gingival margin of the lateral incisor and canine restorations. (C) Twenty-year-old cervical amalgam restorations.



Online Figure 26.4 Surgical access. (A) Class V preparation requiring mucoperiosteal flap reflection with a releasing incision *(arrowhead)*. (B) Completed restoration with suture in place. (C) Suture removed 1 week after the procedure.



Online Figure 26.5 Patients may object to metal restorations that are visible during conversation.



Online Figure 26.6 Restoration for class III tooth preparation using facial approach on mandibular canine. Restoration is 5 years old. (Courtesy of Dr. C. L. Sockwell.)

the surrounding tooth structure. Amalgam restorations are usually easier to finish and polish without damage to the adjacent surfaces.

Disadvantages

The primary disadvantage of class III and V amalgam restorations is that they are metallic and unesthetic. In addition, the preparation for an amalgam restoration typically requires 90-degree cavosurface margins and specific axial depths that allow incorporation of secondary retentive features. These features result in a less conservative preparation than that required for most esthetic restorative materials.

Clinical Technique for Class III Amalgam Restorations

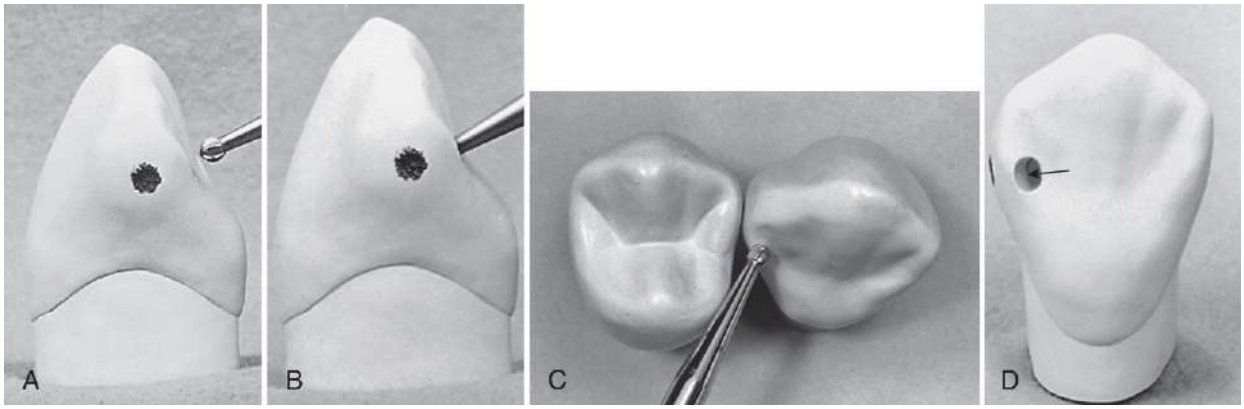
Initial Procedures

After appropriate review of the patient records (including medical history), treatment plan and radiographs, the gingival extension of the preparation should be anticipated. Anaesthesia is usually necessary when a vital tooth is to be restored. Pre-wedging in the gingival embrasure of the proximal site to be restored provides (i) better protection of soft tissue and the rubber dam, (ii) better access because of the slight separation of teeth and (iii) better re-establishment of the proximal contact. The use of a rubber dam is generally recommended; however, cotton roll isolation is acceptable if moisture can be adequately controlled.

Tooth Preparation

A lingual access preparation on the distal surface of the maxillary canine is described here because the use of amalgam in that location is more likely. For esthetic reasons, use of amalgam is best suited for caries that can be accessed from the lingual rather from the facial. A facial approach for a mandibular canine may be indicated, however, if the lesion is more facial than lingual. The mandibular restoration is often not visible at conversational distance (Online Fig. 26.6).

The outline form of the class III amalgam preparation may include only the proximal surface. A lingual dovetail may be indicated if one existed previously or if additional retention is needed for a larger restoration.

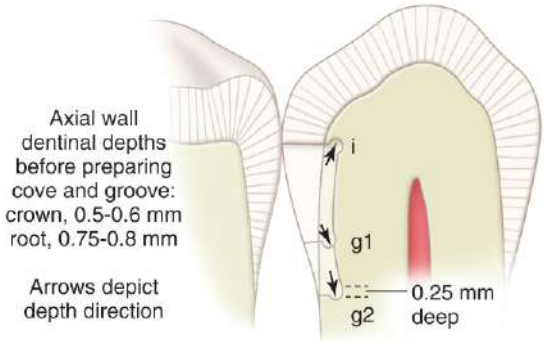


Online Figure 26.7 Entry for class III tooth preparation on maxillary canine. (A) Bur position is perpendicular to the enamel surface at the point of entry. (B) Initial penetration through enamel is directed towards cavitated, caries lesion. (C) Initial entry should isolate the proximal enamel while preserving as much of the marginal ridge as possible. (D) Initial cutting reveals the dentinoenamel junction (DEJ) *(arrow)*.

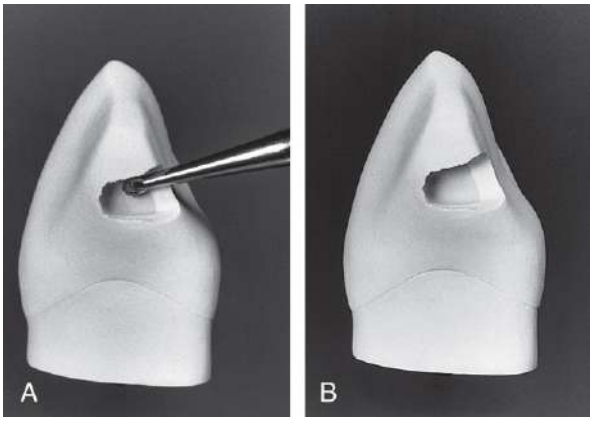
Initial Tooth Preparation

Bur size selection depends on the anticipated size of the lesion. Bur options may include a No. 2 (or smaller) round bur or No. 330 bur. The bur is positioned so that the entry cut penetrates into the caries lesion, which is usually apical to (and slightly into) the contact area. Ideally, the bur is positioned so that its long axis is perpendicular to the lingual surface of the tooth, but directed at a mesial angle as close to the adjacent tooth as possible. (The bur position may be described as perpendicular to the distolingual line angle of the tooth.) This position conserves the marginal ridge enamel (Online Fig. 26.7A–C). Penetration through enamel positions the bur so that additional cutting isolates the proximal enamel affected by caries and removes some or all of the infected dentin. In addition, penetration should be at a limited initial axial depth (i.e. 0.5–0.6 mm) inside the DEJ (Online Fig. 26.7C and D) or at a 0.75–0.8-mm axial depth when the gingival margin is on the root surface (in cementum) (Online Fig. 26.8). This 0.75-mm axial depth on the root surface allows a 0.25-mm distance (the diameter of the No. ¼ bur is 0.5 mm) between the retention groove (which is placed later) and the gingival cavosurface margin. Infected dentin that is deeper than this limited initial axial depth is removed later during final tooth preparation.

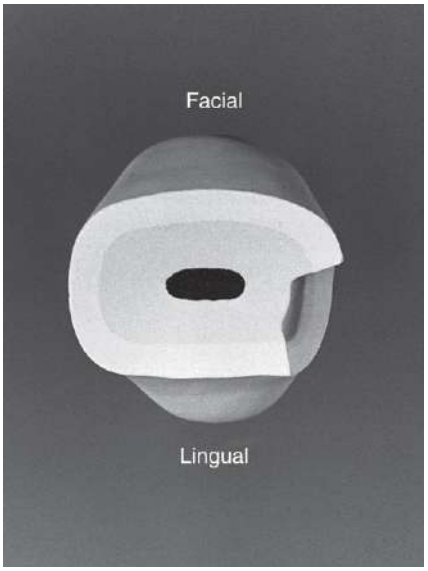
For a small lesion, the facial margin is extended 0.2–0.3 mm into the facial embrasure (if necessary), with a curved outline from the incisal to the gingival margin (resulting in a less visible margin). The lingual outline blends with the incisal and gingival margins in a smooth curve, creating a preparation with little or no lingual wall. The cavosurface angle should be 90 degrees at all margins. The facial, incisal and gingival walls should meet the axial wall at approximately right angles (although the lingual wall meets the axial wall at an obtuse angle or may be continuous with the axial wall) (Online Fig. 26.9). If a large round bur is used, the internal angles are more rounded. The



Online Figure 26.8 Mesiodistal vertical section showing location, depth direction *(arrows)* and direction depth of the retention form in class III tooth preparations of different gingival depths. *i*, Incisal cove; *g1*, gingival groove, enamel margin; *g2*, gingival groove, root surface margin. Distance from outer aspect of *g2* groove to margin is approximately 0.3 mm; bur head diameter is 0.5 mm; direction depth of groove is half this diameter (or approximately 0.3 mm [0.25 mm]).



Online Figure 26.9 Class III tooth preparation on maxillary canine. (A) Round bur shaping the incisal area. The incisal angle remains. (B) Initial shape of the preparation accomplished with a round bur.



Online Figure 26.10 Transverse section of mandibular lateral incisor illustrating that the lingual wall of a class III tooth preparation may meet the axial wall at an obtuse angle and that the axial wall is a uniform depth into dentin and follows the faciolingual contour of the external tooth surface.

axial wall should be uniformly deep into dentin and follow the faciolingual contour of the external tooth surface (Online Fig. 26.10). The initial axial wall depth may be in sound dentin (i.e. shallow lesion), in infected dentin (i.e. moderate to deep lesion) or in existing restorative material, if a restoration is being replaced.

Incisal extension to remove carious tooth structure may eliminate the proximal contact (Online Fig. 26.11). It is important to conserve as much of the distoincisal tooth structure as possible to reduce the risk for subsequent fracture. When possible, it is best to leave the incisal margin in contact with the adjacent tooth.

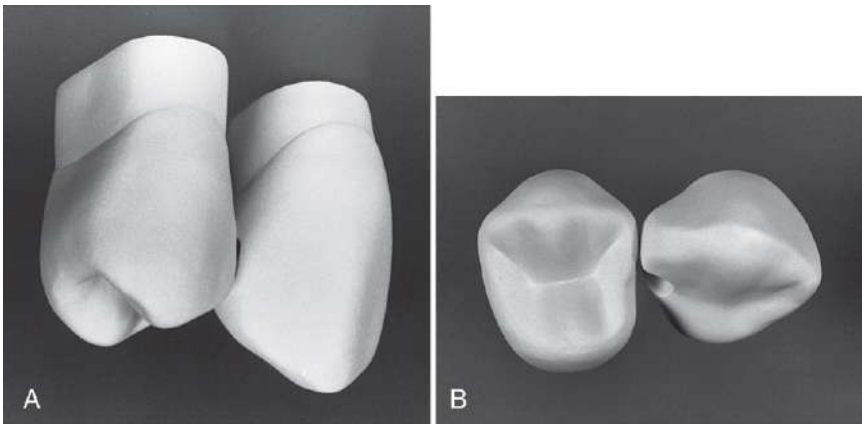
When preparing a gingival wall that is near the level of the rubber dam or apical to it, it is beneficial to place a wedge in the gingival embrasure earlier to depress and protect soft tissue and the rubber dam. As the bur is preparing the gingival wall, it may lightly shave the wedge. A triangular (i.e. anatomic) wedge, rather than a round wedge, is used for a deep gingival margin.

The initial tooth preparation is completed by using a No. ½ round bur to accentuate the axial line angles (Online Fig. 26.12A and B), particularly the axiogingival angle. This facilitates the subsequent placement of retention grooves and leaves the internal line angles slightly rounded. Rounded internal preparation angles permit more complete condensation of the amalgam. The No. ½ round bur may also be used to smooth any roughened, undermined enamel produced at the gingival and facial cavosurface margins (Online Fig. 26.12C). The incisal margin of the minimally extended preparation is often not accessible to the larger round bur without marring the adjacent tooth (Online Fig. 26.12D). Further finishing of the incisal margin is presented later. At this point, the initial tooth preparation is completed.

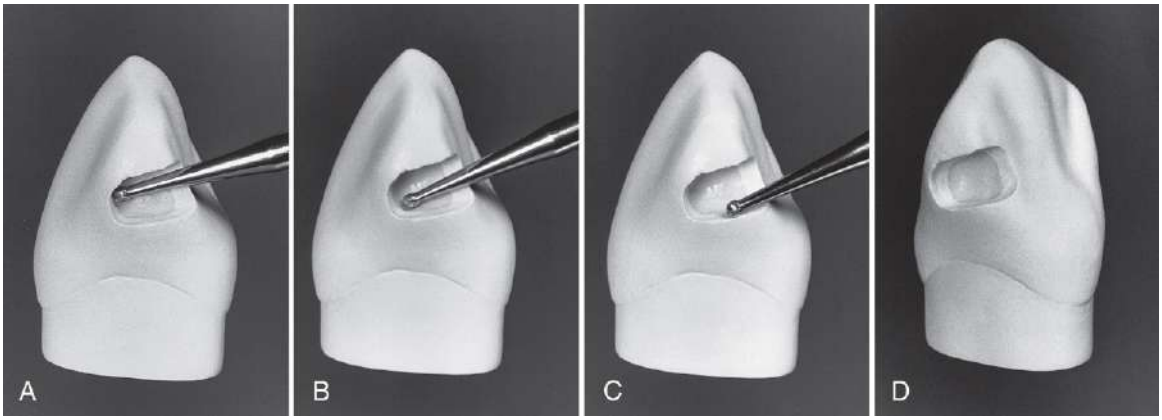
Final Tooth Preparation

Final tooth preparation involves removing any remaining infected dentin; protecting the pulp; developing secondary resistance and retention forms; finishing external walls; and cleaning, inspecting and desensitizing or bonding. Any remaining infected carious dentin on the axial wall is removed by using a slowly revolving round bur (No. 2 or No. 4), appropriate spoon excavators, or both.

For the class III amalgam restoration, resistance form against post-restorative fracture is provided by (i) cavosurface and amalgam margins of 90 degrees, (ii) enamel walls supported by sound dentin, (iii) sufficient bulk of amalgam (minimal 1-mm thickness) and (iv) no sharp preparation internal angles. The box-like preparation form provides primary retention form. Secondary retention form is provided by a gingival groove, an incisal cove and sometimes a lingual dovetail.



Online Figure 26.11 Distofacial (A) and incisal (B) views of the canine to show the curved proximal outline necessary to preserve the distoincisal corner of the tooth. The incisal margin of this preparation example is located slightly incisally of the proximal contact (but whenever possible, the margin may be in the contact area).



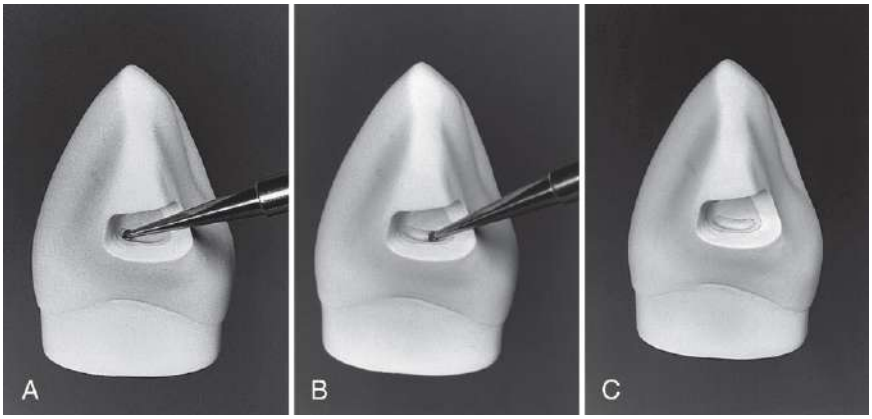
Online Figure 26.12 Refining proximal portion. (A–C) A small, round bur is used to shape the preparation walls, define line angles and initiate removal of any undermined enamel along the gingival and facial margins. (D) Tooth preparation completed, except for the final finishing of the enamel margins and placing the retention form.

The gingival retention groove is prepared by placing a No. ¼ round bur (rotating at low speed) in the axio-facio-gingival point angle. It is positioned in the dentin to maintain 0.2 mm of dentin between the groove and the DEJ. The rotating bur is moved lingually along the axiogingival line angle, with the angle of cutting generally bisecting the angle between the gingival and axial walls. Ideally, the direction of the gingival groove is slightly more gingival than axial (and the direction of an incisal [i.e. occlusal] groove would be slightly more incisal [i.e. occlusal] than axial) (Online Fig. 26.13; see also Fig. 26.8).

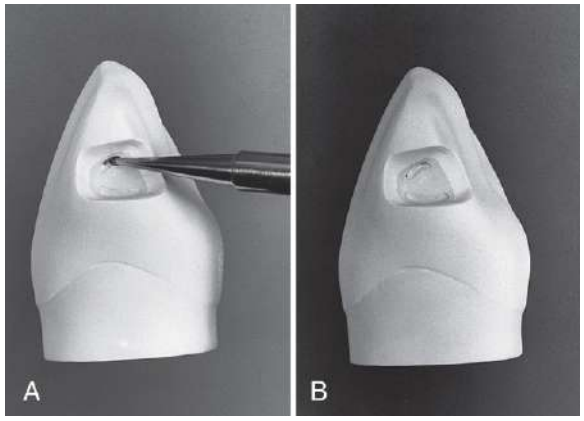
Alternatively, if less retention form is needed, two gingival coves may be used, as opposed to a continuous groove. One each may be placed in the axio-gingivo-facial and axio-gingivo-lingual point angles. The diameter of the ¼ round bur is 0.5 mm, and the depth of the groove should be half this diameter (0.25 mm). (See the location and depthwise direction of the groove, where the gingival wall remains in enamel, in Fig. 26.18.) When preparing a retention groove on the root surface (gingival wall in cementum or dentin), the angle of cutting is more gingival, resulting in the distance from the gingival cavosurface margin to the groove being approximately 0.3 mm (Online Fig. 26.8). Careful technique is necessary in preparing the gingival retention groove. If the dentin that supports gingival enamel is removed, enamel is subject to fracture. In addition, if the groove is placed only in the axial wall, no effective retention form is developed, and a risk of pulpal involvement is possible.

An incisal retention cove is prepared at the axio-facio-incisal point angle with a No. ¼ round bur in dentin, being careful not to undermine enamel. It is directed similarly into the incisal point angle and prepared to half the diameter of the bur (Online Fig. 26.14). Undermining the incisal enamel should be avoided. For the maxillary canine, the palm-and-thumb grasp may be used to direct the bur incisally (Online Fig. 26.15). This completes the typical class III amalgam tooth preparation (Online Fig. 26.16). Similar to class I and II amalgams, it is recommended that the clinician prepares mechanical retention.

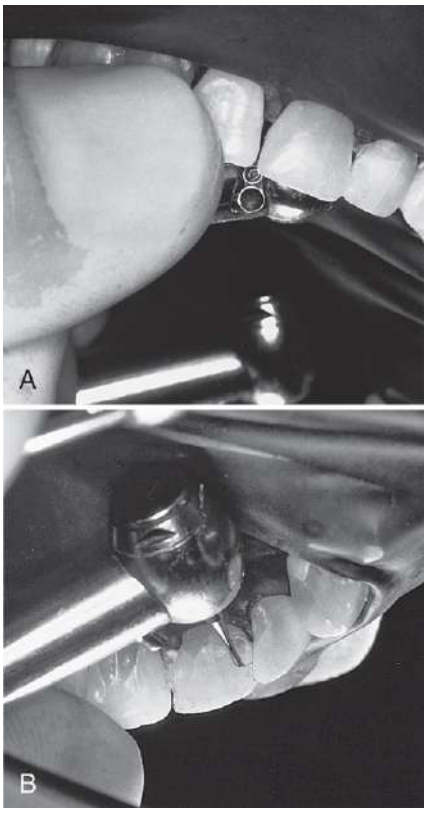
A lingual dovetail is not required in small or moderately sized class III amalgam restorations. It may be used in large preparations, especially preparations with excessive incisal extension in which additional retention form is needed.



Online Figure 26.13 Preparing the gingival retention form. (A) Position of No. ¼ round bur in axio-facio-gingival point angle. (B) Advancing the bur lingually to prepare the groove along the axiogingival line angle. (See Online Fig. 26.8 regarding location, depth direction and direction depth of groove.) (C) Completed gingival retention groove.



Online Figure 26.14 Preparing the incisal retention cove. (A) Position of No. ¼ round bur in the axioincisal point angle. (B) Completed incisal cove.



Online Figure 26.15 Use of the palm-and-thumb grasp to place the incisal retention cove. (A) Hand position showing thumb rest. (B) Hand-piece position for preparing the incisal retention.

The dovetail may not be necessary (even in large preparations), however, if an incisal secondary retention form can be accomplished (Online Fig. 26.17).

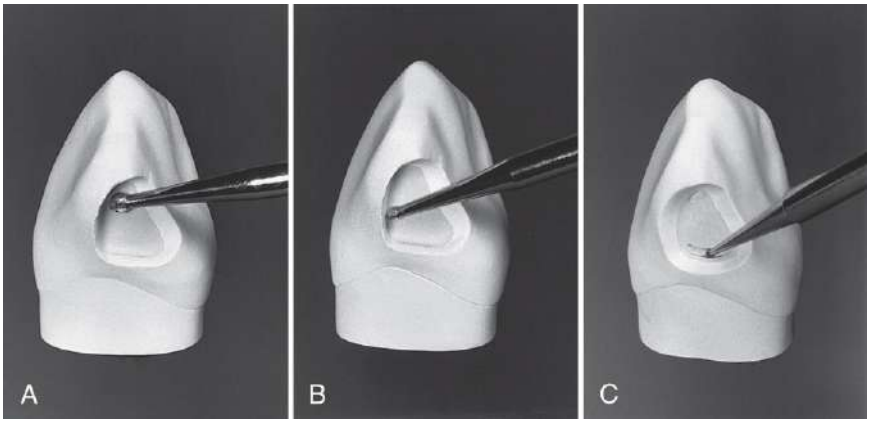
If a lingual dovetail is needed, it is prepared only after initial preparation of the proximal portion has been completed. Otherwise, the tooth structure needed for the isthmus between the proximal portion and the dovetail may be removed when the proximal outline form is prepared. The lingual dovetail should be conservative, generally not extending beyond the mesiodistal midpoint of the lingual



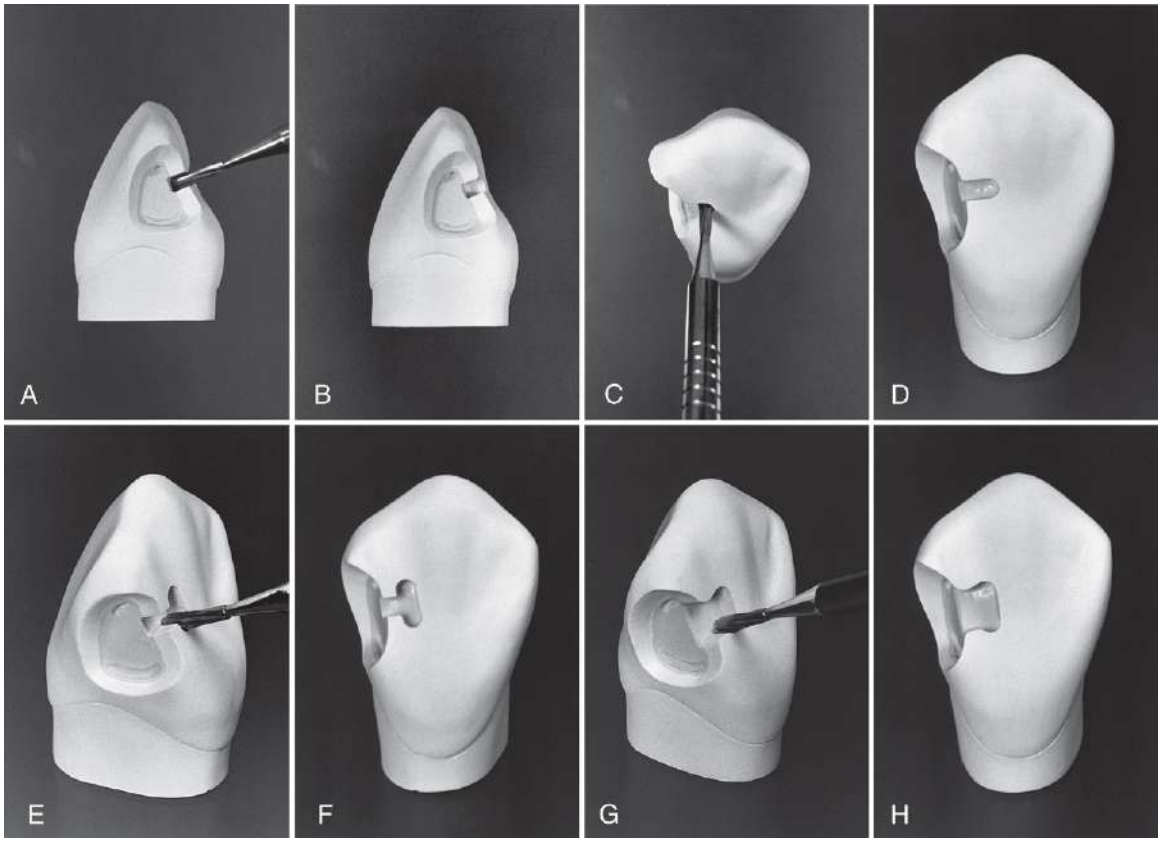
Online Figure 26.16 Completed class III tooth preparation for amalgam restoration.

surface; this varies according to the extent of the proximal caries. The axial depth of the dovetail should approximate 1 mm, and the axial wall should be parallel to the lingual surface of the tooth. This wall may or may not be in dentin. The No. 245 bur is positioned in the proximal portion at the correct depth and angulation and moved in a mesial direction (Online Fig. 26.18A and B). The correct angulation places the long axis of the bur perpendicular to the lingual surface. The bur is moved to the point that corresponds to the most mesial extent of the dovetail (Online Fig. 26.18C and D). The bur is then moved incisally and gingivally to create sufficient incisogingival dimension to the dovetail (approximately 2.5 mm) (Online Fig. 26.18E and F). The incisal and gingival walls of the isthmus are prepared in smooth curves connecting the dovetail to the proximal outline form (Online Fig. 26.18G and H).

The gingival margin trimmer can be used to bevel (or round) the axiopulpal line angle (i.e. the junction of the proximal and dovetail preparation). This increases the strength of the restoration at the junction of the proximal and lingual portions by providing bulk and reducing stress concentration. The lingual convergence of the dovetail's external walls (prepared with the No. 245 bur) usually provides a sufficient retention form. Retention coves, one in the incisal corner and one in the gingival corner (Online Fig. 26.19), may be placed in the dovetail to enhance retention if the axial wall of the dovetail is in dentin. The coves are prepared with the No. ¼ round bur in dentin that does not immediately support the lingual enamel. This preparation may require deepening of the axial wall. Unsupported enamel is removed, the walls or margins are smoothed and the cavosurface angles are refined, where indicated. The 8-3-22 hoe is recommended for finishing minimally extended margins (Online Fig. 26.20). If the gingival margin is in enamel, a slight bevel (approximately 20 degrees) is necessary to ensure full-length enamel rods forming the cavosurface margin. All the walls of the preparation should meet the external tooth surface to form a right angle (i.e. butt joint) (Online Fig. 26.21; see also Fig. 26.16). The various steps involved



Online Figure 26.17 Extensive class III tooth preparation. (A) Initial tooth preparation with No. 2 round bur. (B) Defining line angles and removing undermined enamel with No. ½ round bur. (C) Placing the retention groove using No. ¼ round bur. Note the completed incisal cove.



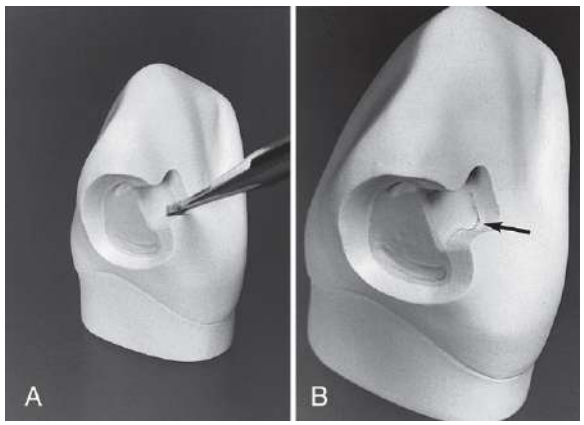
Online Figure 26.18 Lingual dovetail providing additional retention for extensive amalgam restoration. (A) Bur position at correct depth and angulation to begin cutting. (B) Initial cut in beginning dovetail. (C) Bur moved to most mesial extent of dovetail. (D) If possible, cutting should not extend beyond the midlingual position. (E) Bur cutting gingival extension of the dovetail. (F) Incisal and gingival extensions of the dovetail. (G) Completing the isthmus. The proximal and lingual portions are connected by the incisal and gingival walls in smooth curves. (H) Completed lingual dovetail.

in the clinical procedure with the dovetail are shown in Online Fig. 26.22. The completed tooth preparation should be cleaned of any residual debris and inspected. Careful assessment should be made to see that all of the caries has been removed, that the depth and retention are appropriate and that cavosurface margins provide for the amalgam bulk.

Restorative Technique

Desensitizer Placement

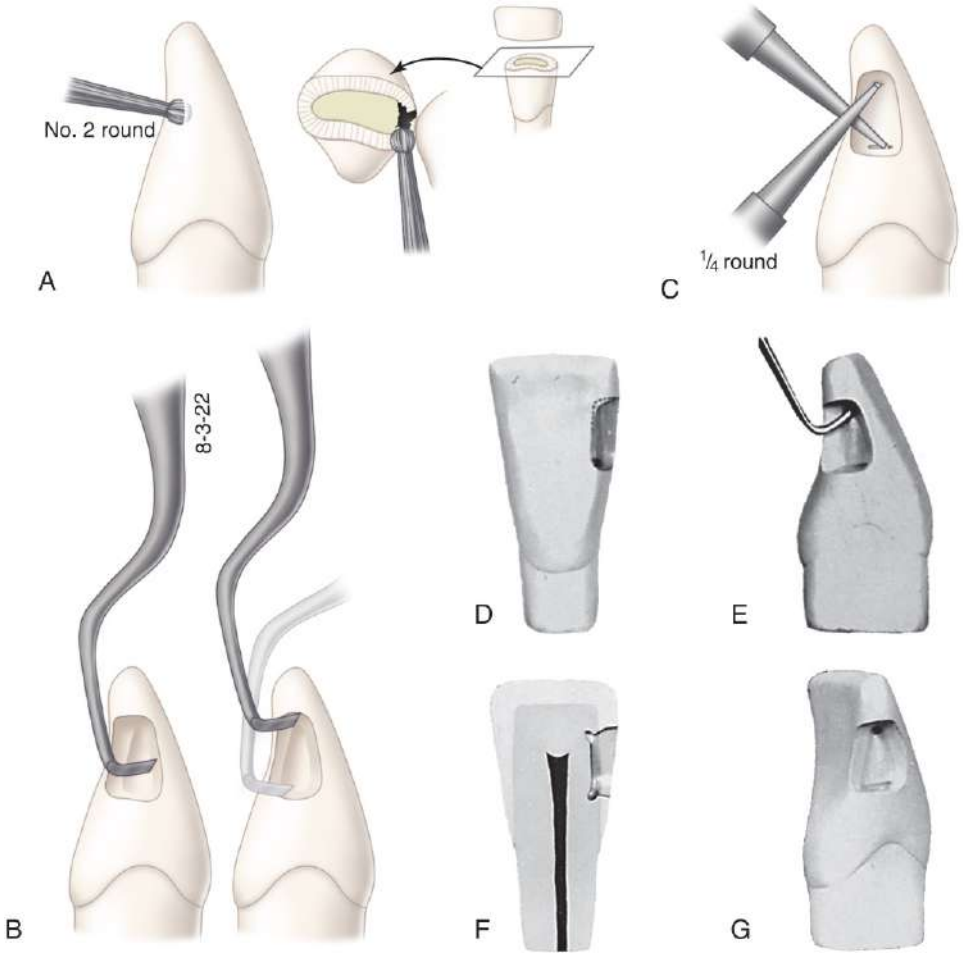
The use of a dentin desensitizer over the prepared tooth structure before placing amalgam is generally recommended. The dentin desensitizer is rubbed onto the prepared



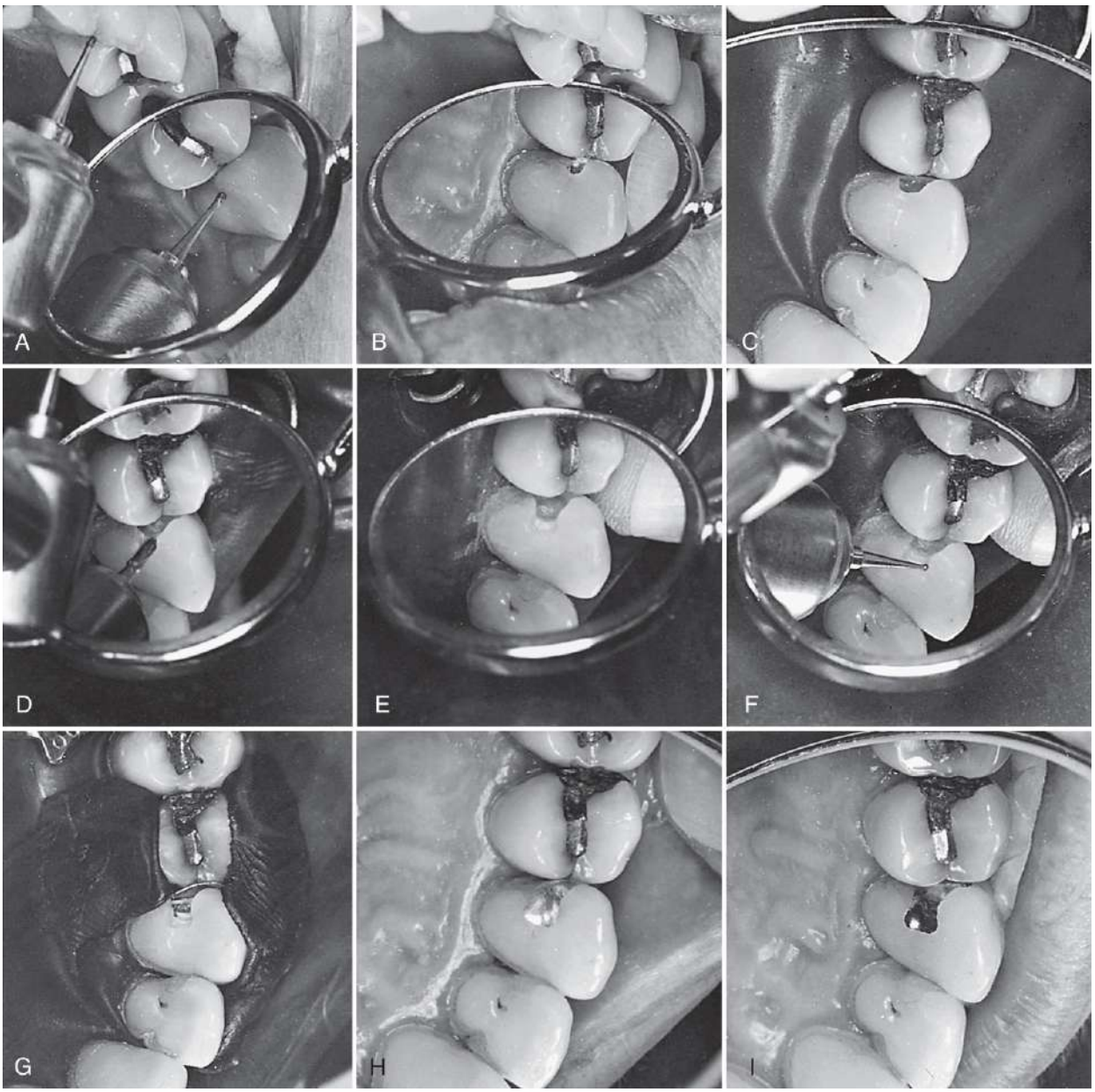
Online Figure 26.19 Ensuring retention in lingual dovetail (often optional). (A) Position of No. 33½ bur for cutting the retention cove. (B) Preparation of the cove should not remove the dentinal support of the lingual enamel *(arrow)*.



Online Figure 26.21 Completed distolingual class III tooth preparation for amalgam.



Online Figure 26.20 Class III tooth preparation for amalgam restoration on the mandibular incisor. (A) Entering the tooth from the lingual approach. (B) Finishing the facial, incisal and gingival enamel margins with an 8-3-22 triple angle hoe. Note how the reverse bevel blade is used on the gingival enamel. (C) Placing incisal and gingival retention forms with No. ¼ round bur. (D) *Dotted line* indicates the outline of the additional extension that is sometimes necessary for access in placing the incisal retention cove. (E) Position of a bi-bevelled hatchet to place the incisal retention cove. (F) The axial wall forms a convex surface over the pulp. (G) Completed tooth preparation. Note the gingival retention groove.

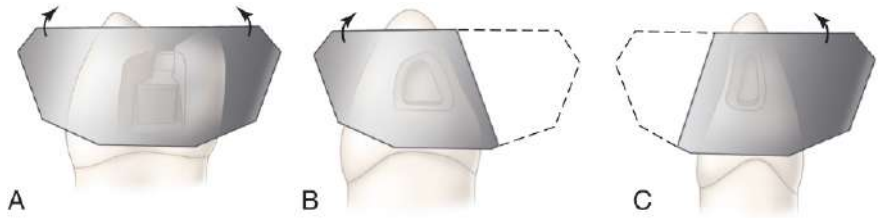


Online Figure 26.22 Distolingual tooth preparation and restoration. (A) Bur position for entry. (B) Penetration made through the lingual enamel to the caries. (C) Proximal portion completed, except for the retention form. (D) Preparing the dovetail. (E) Completed preparation, except for the retention groove and the coves. (F) Bur position for the incisal cove in the dovetail. (G) Rigid material-supported matrix ready for the insertion of amalgam. (H) Carving completed and rubber dam removed. (I) Polished restoration.

tooth surface for 30 s and excess moisture is removed without desiccating the dentin.

Matrix Placement

The wedged, rigid material-supported sectional matrix may be used for the class III amalgam restoration. Insertion of amalgam into the class III tooth preparation is usually from the lingual aspect. It is essential to trim the lingual portion of the sectional matrix material correctly to avoid covering the preparation and blocking access for insertion of the amalgam. A length of 5y16 inch (8 mm) wide, 0.002 inch (0.05 mm) thick stainless steel matrix material that covers one-third of the facial surface and extends through the proximal to the lingual surface is obtained. The lingual portion is trimmed at an angle that corresponds approximately to the slope of the lingual surface of the tooth (Online Fig. 26.23). The section of matrix is burnished on a resilient paper pad to create the desired contact and contour form. The sectional matrix is placed in position and wedged from the facial or lingual embrasure, whichever is greater. The facial and lingual portions of the matrix are stabilized with a rigid material (Online Fig. 26.22G). Precontoured metallic matrices may be used (instead of custom-made matrices) if the contour of the precontoured matrix coincides with that of the proximal surface being



Online Figure 26.23 Matrix strip design. (A) Design required for rigid material-supported matrix for class II tooth preparations. (B) Alteration necessary for class III preparation on the maxillary canine. (C) Alteration necessary for the mandibular incisor. The strip material is cut to approximate the slope of the lingual surface.

restored. If the preparation is small and the matrix is sufficiently rigid, additional rigid material to support the matrix may not be required.

Condensation and Carving

Insertion of the amalgam, initial carving, matrix removal, wedge removal and final carving are similar to the techniques for posterior teeth. When properly placed, conservative restorations in incisors and canines are relatively inconspicuous (Online Fig. 26.24). Online Fig. 26.25 illustrates a class III amalgam restoration in a mandibular incisor.

Finishing and Polishing

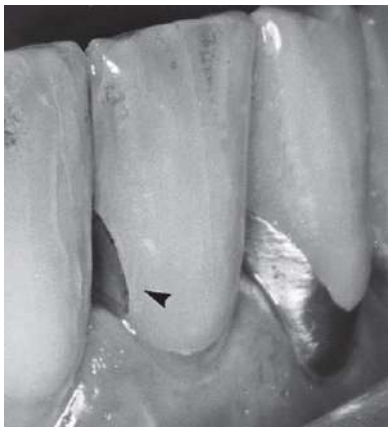
The finishing and polishing techniques and procedures are the same as those presented in Chapter 13.

Clinical Technique for Class V Amalgam Restorations

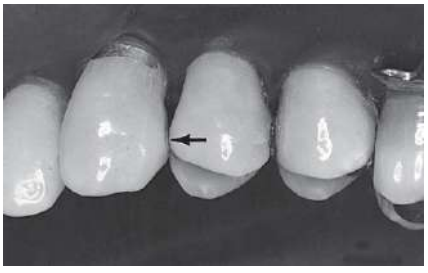
Initial Procedures

Proper isolation prevents moisture contamination of the operating site, enhances asepsis and facilitates access and visibility. Moisture in the form of saliva, gingival sulcular fluid or gingival haemorrhage must be excluded during caries removal, liner application, and insertion and carving of amalgam. Moisture impairs visual assessment, may contaminate the pulp during caries removal (especially with a pulpal exposure) and negatively affects the physical properties of restorative materials. The gingival margin of class V tooth preparations is often apical to the gingival crest. Such a gingival margin necessitates retraction of the free gingiva with a retraction cord or appropriate rubber dam and retainer to protect it and to provide access while eliminating seepage of sulcular fluid into the tooth preparation or restorative materials.

These isolation objectives are met by local anaesthesia and isolation by (i) a cotton roll and a retraction cord or (ii) a rubber dam and a suitable cervical retainer (Online Fig. 26.26). Isolation with a cotton roll and a retraction cord is satisfactory when properly performed. This type of isolation is practical and probably the approach most often used. The retraction cord should be placed in the sulcus before initial tooth preparation to reduce the possibility of cutting instruments damaging the free gingiva. The cord



Online Figure 26.25 Class III amalgam restoration on the mandibular incisor *(arrowhead)*.



Online Figure 26.24 Inconspicuous facial margin *(arrow)* of class III amalgam restoration on the maxillary canine.



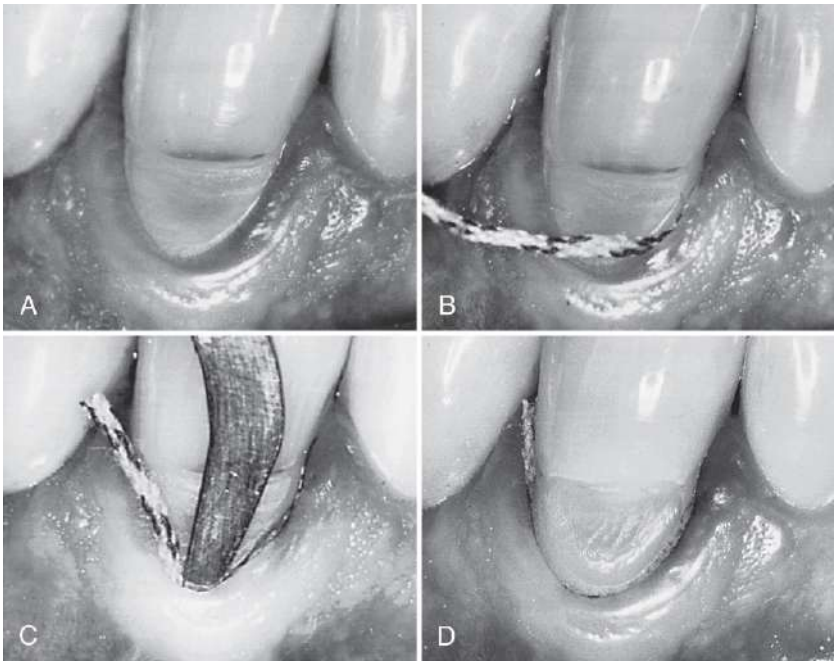
Online Figure 26.26 A rubber dam and a No. 212 retainer may be required to isolate the carious area properly.

should produce a temporary, adequate, a traumatic apical retraction and lateral deflection of the free gingiva. The cord may be treated with haemostatic preparations containing aluminium chloride or ferrous sulphate. Alternatively, the cord may be treated with epinephrine. However, caution must be used as epinephrine on abraded gingiva can be absorbed rapidly into the circulatory system, causing an increase in blood pressure, elevated heart rate and possible dysrhythmia. The retraction cord may be braided, twisted or woven. The diameter of the cord should be easily accommodated in the gingival sulcus. An appropriate amount is cut to a length ¼ inch (6 mm) longer than the gingival margin. Some operators prefer to place the cord in a Dappen dish, wet it with a drop of haemostatic solution and blot it with a 2 × 2 inch (5 × 5 cm) gauze to remove excess liquid. A braided or woven cord is usually easier to use because it does not unravel during placement. A larger cord can be inserted over the first cord if the sulcus is large enough to accommodate two cords. A thin, blunt-edged instrument blade or the side of an explorer tine may be used to gently insert the cord progressively into place. A slight backward direction of the instrument as it steps along the cord helps prevent dislodgment of previously inserted cord (Online Fig. 26.27). In addition, using a second instrument stepping along behind the first instrument can help prevent dislodgment of cord. In addition, using the air syringe or cotton pellets to reduce or absorb the sulcular fluid in the cord already placed is helpful during cord placement. The cord results in adequate retraction in a short time. If significant blanching of the free gingiva is observed (or if too much pressure has to be applied to place the cord), it means that an oversized cord has been selected, and it should therefore be exchanged with a cord of smaller diameter. The cord can be moistened before or after placement with the haemostatic solution if slight haemorrhage is anticipated or observed. Alternatively, the cord can be used dry.

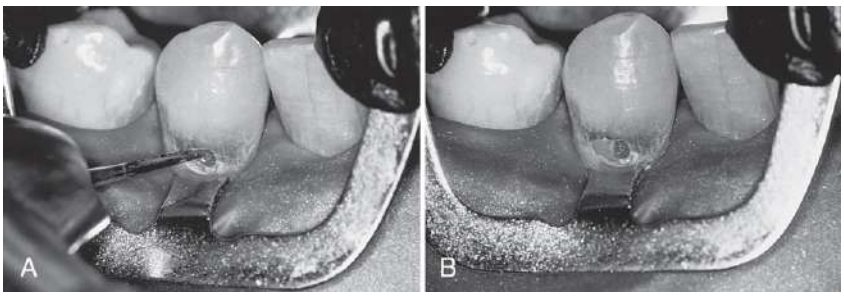
The cord usually remains in place throughout tooth preparation as well as insertion and initial carving of amalgam. While carving amalgam at the gingival margin, the presence of the cord may cause difficulty in feeling the unprepared tooth surface apical to the margin to prevent undercarving of the margin; this results in overcontouring and marginal excess. In this instance, after carving gross excess, the cord can be teased from its place before the carving is completed.

Tooth Preparation

Proper outline form for class V amalgam tooth preparations results in extending the cavosurface margins to sound tooth structure while maintaining a limited axial depth of 0.5 mm inside the DEJ and 0.75 mm inside cementum (when on the root surface). The outline form for the class V amalgam tooth preparation is determined primarily by the location and size of the caries or old restorative material. Clinical judgement determines final preparation outline, especially when the cavosurface margins approach or extend into areas of enamel decalcification. The operator must observe the prepared enamel wall to evaluate the depth of the decalcified enamel and to determine if cavitation exists peripheral to the wall. When no cavitation has occurred and when the decalcification does not extend appreciably into the enamel, extension of the outline form often should cease. In some cases, if all decalcification



Online Figure 26.27 Use of the retraction cord for isolation of a class V lesion. (A) Pre-operative view. (B) Cord placement initiated. (C) Cord placement using a thin, flat-bladed instrument. (D) Cord placement completed.

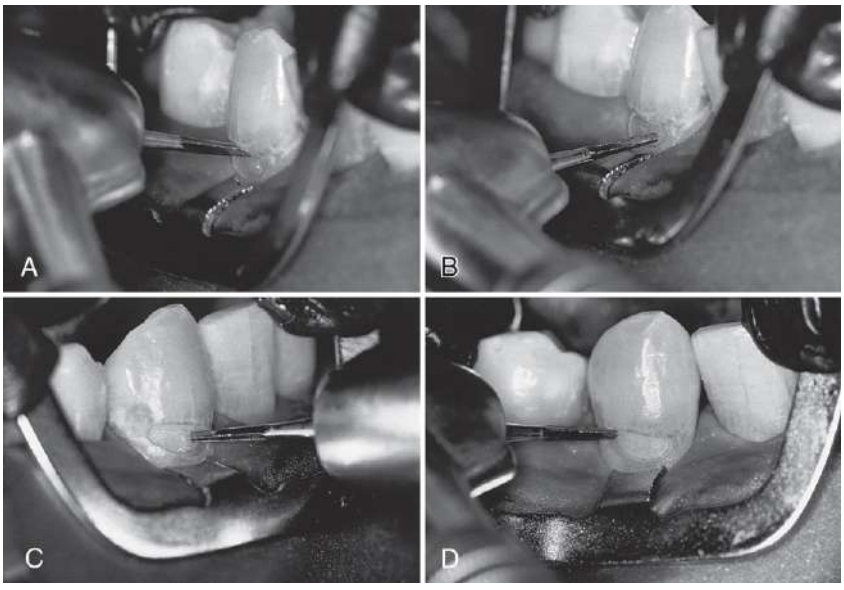


Online Figure 26.28 Starting class V tooth preparation. (A) Bur positioned for entry into caries lesion. (B) The entry cut is the beginning of the outline form having a limited axial depth. (The end of the bur in the centre of the lesion may be in the carious tooth structure or in the air.)

were included in the outline form, the preparation would extend into the proximal cervical areas (if not circumferentially around the tooth). Such a preparation would be difficult and perhaps unrestorable. A full-coverage restoration should be considered for teeth with extensive cervical decalcification.

Initial Tooth Preparation

A class V amalgam restoration is not used often in a mandibular canine, but it is presented here for illustration. The same general principles for tooth preparation apply for all other tooth locations. A tapered fissure bur of suitable size (e.g. No. 271) is used to enter the caries lesion (or existing restoration) to a limited initial axial depth of 0.5 mm inside the DEJ (Online Fig. 26.28). This depth is usually 1–1.25 mm total axial depth, depending on the incisogingival (i.e. occlusogingival) location. The enamel is considerably thicker occlusally and incisally than cervically. If the preparation is on the root surface, however, the axial depth is approximately 0.75 mm. The end of the bur at the initial depth is in dentin, in infected carious dentin or in old restorative material. The edge of the end of the bur can be used to penetrate the area; this is more efficient than using the flat end of the bur, reducing the possibility of the bur's 'crawling'. When the entry is made, the bur orientation is adjusted to ensure that all external walls are perpendicular to the external tooth surface and parallel to the enamel rods (Online Fig. 26.29). Often, this requires changing the orientation of the hand-piece to accommodate the cervical mesiodistal and incisogingival (i.e. occlusogingival) convexity of the tooth. The preparation is extended incisally, gingivally, mesially and distally until the cavosurface margins are positioned in sound tooth structure such that an initial axial depth of 0.5 mm inside the DEJ (if on the root surface, the axial depth is 0.75 mm) is established. When extending mesially and distally, it may be necessary to protect the rubber dam from the bur by placing a flat-bladed instrument over the dam (Online Fig. 26.30). Because the axial wall follows the mesiodistal and incisogingival (i.e. occlusogingival) contours of the facial surface of the tooth, it is usually convex in both directions. In addition, the axial wall usually is slightly deeper at the incisal wall, where more enamel (i.e. approximately 1–1.25 mm in depth) is



Online Figure 26.29 When extending incisally (A), gingivally (B), mesially (C) and distally (D), the bur is positioned to prepare these walls perpendicular to the external tooth surface.



Online Figure 26.30 The flat-bladed instrument protects the rubber dam from the bur.

present than at the gingival wall, where little or no enamel (i.e. approximately 0.75–1 mm in depth) may be present. A depth of 0.5 mm inside the DEJ permits placement of necessary retention grooves without undermining enamel. This subtle difference in depth also serves to increase the thickness of the remaining dentin (between the axial wall and the pulp) in the gingival aspect of the preparation to aid in protecting the pulp. For the tooth preparation that is extended incisogingivally, the axial wall should be more convex (because it follows the contour of the DEJ).

Alternatively, an appropriate carbide bur (usually No. 2 or No. 4) may be used for the initial tooth preparation. Round burs are indicated in areas inaccessible to a fissure bur that is held perpendicular to the tooth surface. If needed, smaller round burs may also be used to define the internal angles in these preparations, enhancing proper placement of the retention grooves.

Final Tooth Preparation

Final tooth preparation involves removing any remaining infected dentin; pulp protection; retention form; finishing external walls; and cleaning, inspecting and desensitizing. Any remaining infected axial wall dentin is removed with a No. 2 or No. 4 round bur. Any old restorative material (including base and liner) remaining may be left if (i) no clinical or radiographic evidence of recurrent caries exists, (ii) the periphery of the base and liner is intact and (iii) the tooth is asymptomatic. With proper outline form, the axial line angles are already in sound dentin. If needed, an appropriate liner or base is applied.

Because the mesial, distal, gingival and incisal walls of the tooth preparation are perpendicular to the external tooth surface, they usually diverge facially. Consequently, this form provides no inherent retention, and retention form must be provided because the primary retention form for an amalgam restoration is macromechanical. A No. ¼ round bur is used to prepare two retention grooves, one along the incisoaxial line angle and the other along the gingivoaxial line angle (Online Fig. 26.31). The handpiece is positioned so that the No. ¼ round bur is directed generally to bisect the angle formed at the junction of the axial wall and the incisal (i.e. occlusal) wall. Ideally, the direction of the incisal (i.e. occlusal) groove is slightly more incisal (i.e. occlusal) than axial, and the direction of the gingival groove is slightly more gingival than axial. Alternatively, four retention coves may be prepared, one in each of the four axial point angles of the preparation (Online Fig. 26.32).

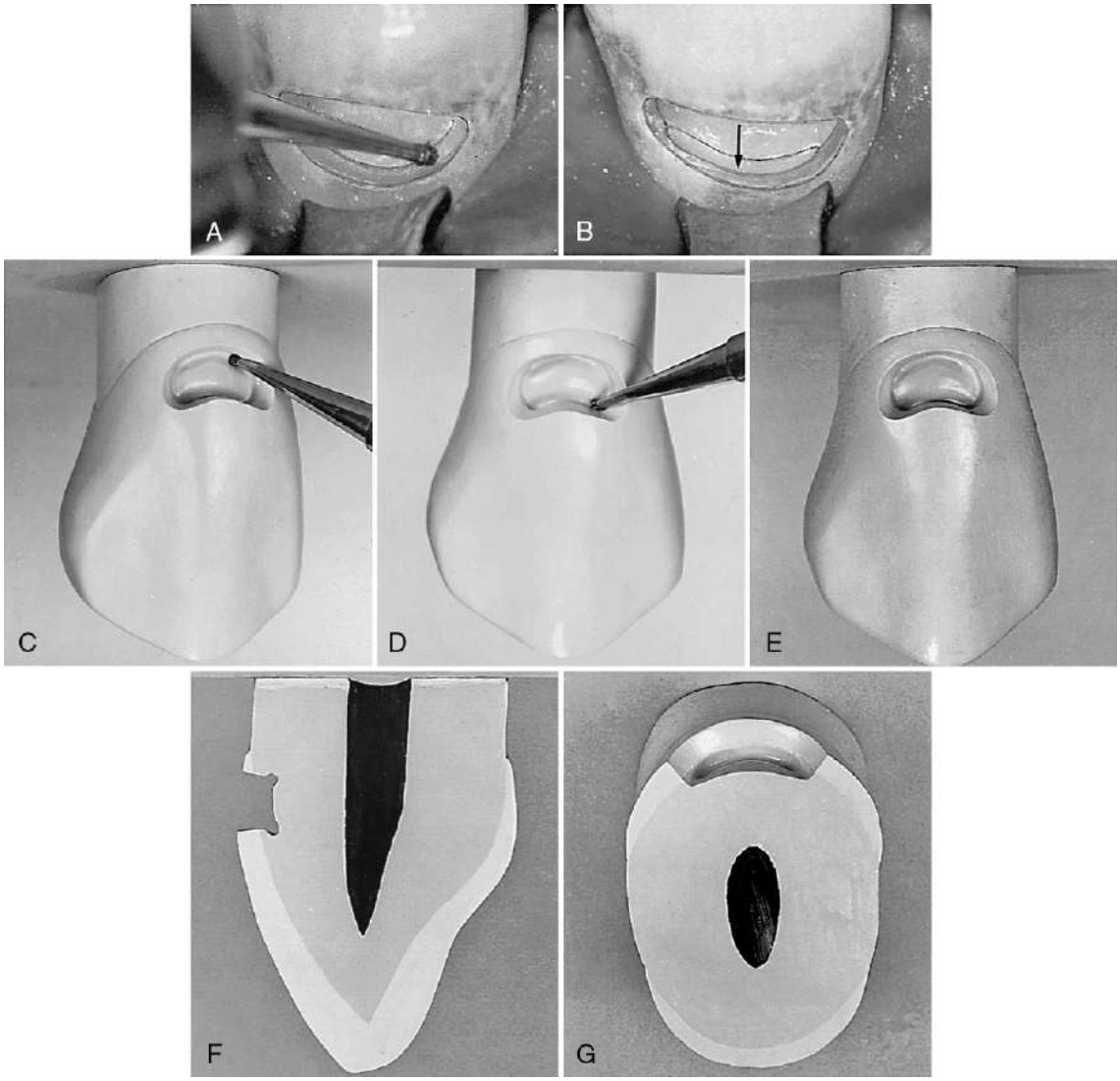
Using four coves instead of two full-length grooves conserves the dentin near the pulp, reducing the possibility of a mechanical pulp exposure. The depth of the grooves should be approximately 0.25 mm, which is half the diameter of the bur. It is important that the retention grooves be adequate because they provide the only retention form to the preparation. Regardless, the grooves should not remove dentin immediately supporting enamel. In a large class V amalgam preparation, extending the retention groove circumferentially around all the internal line angles of the tooth preparation may enhance the retention form.

If access is inadequate for use of the No. ¼ round bur, an angle-former chisel may be used to prepare the retention form. In addition, a No. 33½ bur can be used. Both methods result in retention grooves that are angular, but positioned in the same location and approximately to the same depth as when the No. ¼ round bur is used. The rounded retention form placed with the No. ¼ round bur is generally preferred, however, because amalgam can be condensed into rounded areas better than into sharp areas, resulting in better adaptation of amalgam into the retention grooves. If necessary, suitable hand instruments (e.g. chisels, margin trimmers) are used to plane the enamel margins, verifying soundness and 90-degree cavosurface angles. Finally, the preparation is cleaned and inspected for completeness. A desensitizer is then applied.

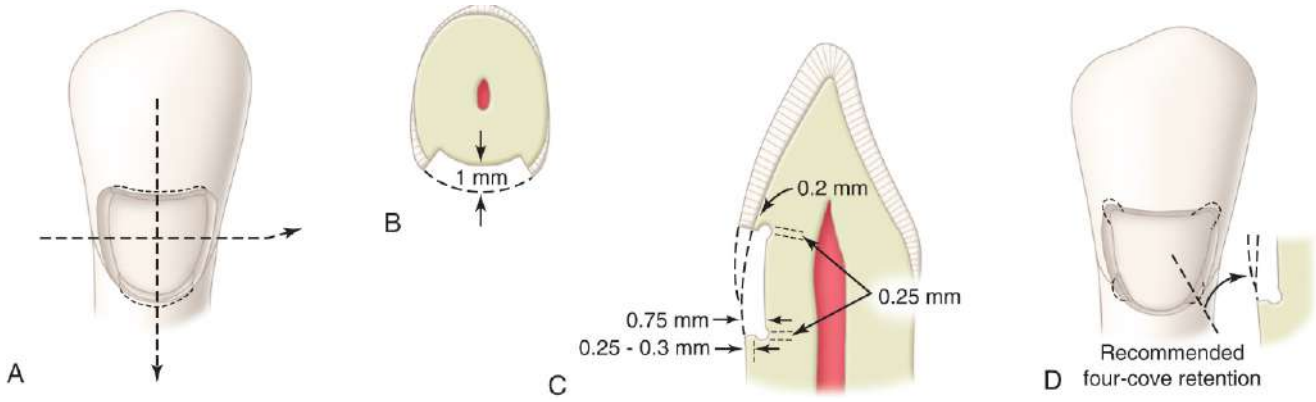
Large Preparations That Include Line Angles

Caries on the facial (or lingual) surface may extend beyond the line angles of the tooth. Maxillary molars, particularly the second molars, are most commonly affected by these extensive defects (Online Fig. 26.33A). In this example, if the remainder of the distal surface is sound and the distal caries is accessible facially, the facial restoration should extend around the line angle. This prevents the need for a class II proximal restoration to restore the distal surface. As much of the preparation as possible should be completed with a fissure bur. A round bur, approximately the same diameter as the fissure bur, is then used to initiate the distal portion of the preparation (Online Fig. 26.33B and C). Smaller round burs should be used to accentuate the internal line angles of the distal portion. Preparing the facial portion first provides better access and visibility to the distal portion. Occasionally, hand instruments may be useful for completing the distal half of the preparation when space for the handpiece is limited (Online Fig. 26.33D–F).

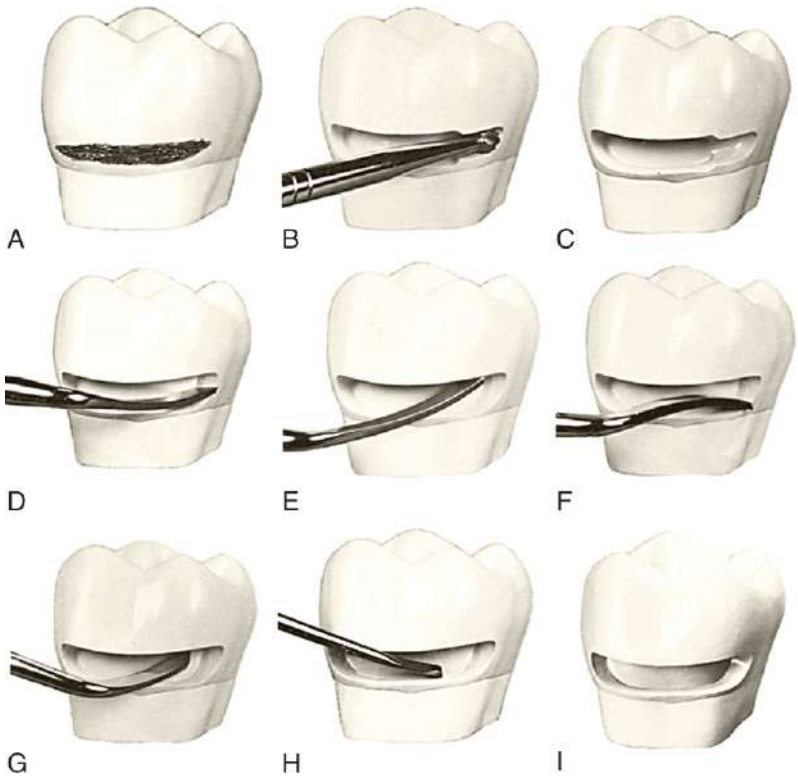
Grooves placed along the entire length of the occlusoaxial and gingivoaxial line angles help ensure retention of



Online Figure 26.31 Retention form. (A) A No. ¼ round bur positioned to prepare the gingival retention groove. (B) Gingival retention groove *(arrow)* prepared along the gingivoaxial line angle generally to bisect the angle formed by the gingival and axial walls. Ideally, the direction of preparation is slightly more gingival than pulpal. An incisal retention groove is prepared along the incisoaxial line angle and directed similarly. (C, D) A groove is placed with a No. ¼ round bur along the gingivoaxial and incisoaxial line angles 0.2 mm inside the dentinoenamel junction (DEJ) and 0.25 mm deep. Note the slight pulpal inclination of the shank of the No. ¼ round bur. (E) Facial view. (F) Incisogingival section. Grooves depthwise are directed mostly incisally (gingivally) and slightly pulpally. (G) Mesiodistal section.



Online Figure 26.32 (A–C) Extended class V tooth preparation (A) with the axial wall contoured parallel to the dentinoenamel junction (DEJ) mesiodistally (B) and incisogingivally (C). The axial wall pulpal depth is 1 mm in the crown and 0.75 mm in the root. In addition, note location and direction depth (0.25 mm) of the retention grooves and the dimension of the gingival wall (0.25 mm) from the root surface to the retention groove. (D) Large class V preparation with retention coves prepared in the four axial point angles.



Online Figure 26.33 Tooth preparation on maxillary molar. (A) Caries extending around distofacial corner of the tooth. (B, C) Distal extension is accomplished with round bur. (D–F) A gingival margin trimmer may be useful in completing the distal half of the preparation when hand-piece access is limited. (G) A gingival margin trimmer may be used to provide the retention grooves. (H) An angle-former chisel may be used to prepare the retention grooves in the distal portion of the preparation. (I) Completed tooth preparation.

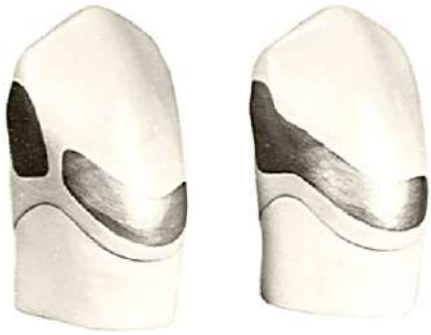
the restoration. The No. ¼ round bur is used as previously described to prepare the retention grooves. A gingival margin trimmer or a 7-85-2½-6 angle-former chisel can be used in the distal half of the preparation to provide retention form when access for the handpiece is limited (Online Fig. 26.33G and H).

Because of the proximity of the coronoid process, access to the facial surfaces of maxillary molars, particularly the second molars, is often limited. Having the patient partially closes and shifts the mandible towards the tooth being restored improves access and visibility (Online Fig. 26.34).

If the class V outline form approaches an existing proximal restoration, it is better to extend slightly into the bulk of the proximal restoration, rather than to leave a thin section of the tooth structure between the two restorations (Online Fig. 26.35). In this illustration, the previously placed amalgam served as the distal wall of the preparation. When proper treatment requires class II and V amalgam restorations on the same tooth, the class II preparation and restoration is completed before initiating the class V restoration. If the class V restoration were done first, it might be damaged by the matrix band and wedge needed for the class II restoration.



Online Figure 26.34 The mandible shifted laterally for improved access and visibility.



Online Figure 26.35 When a class V outline form closely approaches an existing restoration, the preparation should be extended to remove the remaining thin enamel wall to achieve adjoining restorations.

Restorative Technique

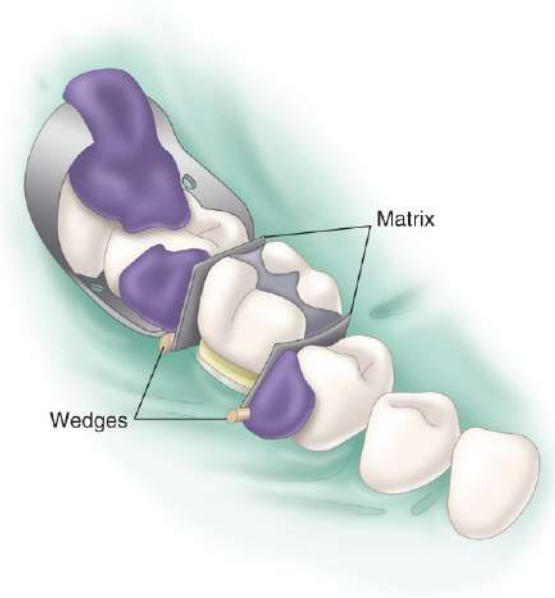
Desensitizer Placement

The same considerations presented earlier apply for the class V amalgam restoration.

Matrix Placement

Most class V amalgam restorations are placed without the use of any type of matrix. The most difficult condensation occurs in a tooth preparation with an axial wall that is convex mesiodistally. Two alternative methods for insertion may be used. The preferred method is the application of a matrix that confines amalgam in the mesial and distal portions of the preparation (Online Fig. 26.36). Short lengths of stainless steel matrix material, one each for the mesial and distal surfaces, are passed through the proximal contacts, carefully guided into the gingival sulcus and wedged. The strips must be wide enough to extend occlusally through the respective proximal contacts and long enough to extend slightly past the facial line angles. The strip usually requires rigid material support for stability. The strips offer resistance against condensing the mesial and distal portions, which provides support for condensing the centre of the restoration. The gingival edge of the steel strip often must be trimmed to conform to the circumferential contour (level) of the base of the gingival sulcus to prevent soft tissue damage. Rather than using two short pieces, the operator can use a longer length that may be passed through one proximal contact, extended around the lingual surface and passed through the other contact, forming a U-shaped matrix. Trimming the gingival edge to conform to the interproximal soft tissue anatomy usually is more difficult with one matrix strip than when two strips are used.

A conventional Tofflemire band and retainer may be used with a window cut into the band allowing access to the preparation for condensation (Online Fig. 26.37). Alternatively, the tooth may be prepared and restored in sections without using a matrix. Each successive section of the preparation should be extended slightly into the previously condensed portion to ensure caries removal. This procedure is time-consuming but effective.

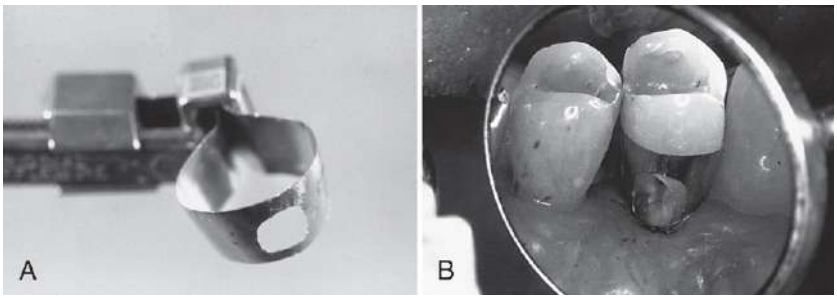


Online Figure 26.36 Application of the matrix to confine amalgam in the mesial and distal extensions of the preparation.

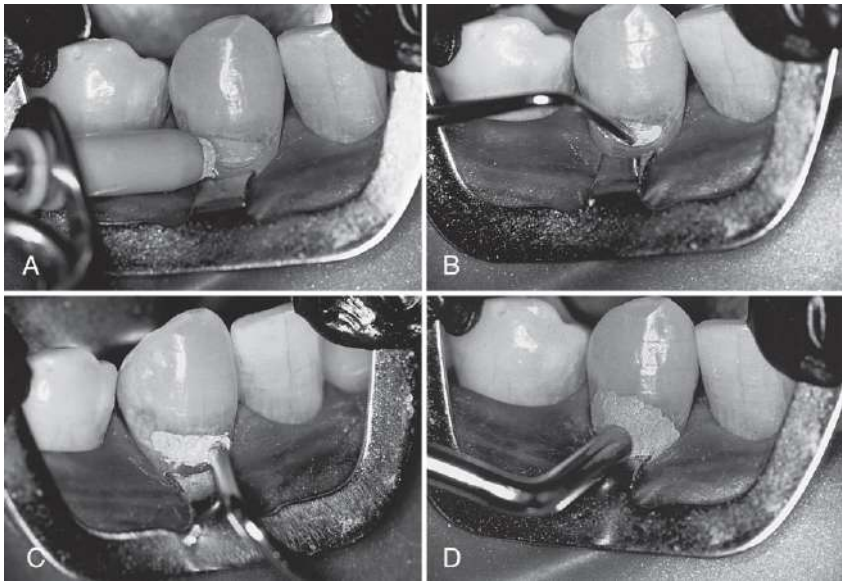
Insertion and Carving of the Amalgam

The amalgam carrier is used to insert the mixed amalgam into the preparation in small increments (Online Fig. 26.38A). Amalgam is condensed into the retention areas first by using an appropriate condenser (Online Fig. 26.38B). Next, amalgam is condensed against the mesial and distal walls of the preparation (Online Fig. 26.38C). Finally, a sufficient bulk of amalgam is placed in the central portion to allow for carving the correct contour (Online Fig. 26.38D). As the surface of the restoration becomes more convex, condensation becomes increasingly difficult. It is important to guard against the 'landsliding' of amalgam during overpacking. A large condenser or flat-bladed instrument held against amalgam may help resist pressure applied elsewhere on the restoration (Online Fig. 26.39).

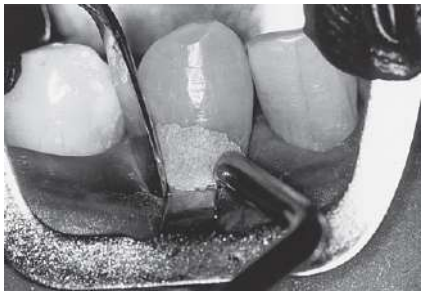
Carving may begin immediately after the insertion of amalgam (Online Fig. 26.40). All carving should be done using the side of the explorer tine or a Hollenback No. 3 carver held parallel to the margins. The side of the carving



Online Figure 26.37 Customized matrix band used to restore an area of proximal root caries. (A) Conventional Tofflemire matrix with window cut into the band to allow access for condensation. (B) Matrix in place around the tooth, allowing lingual access to preparation.

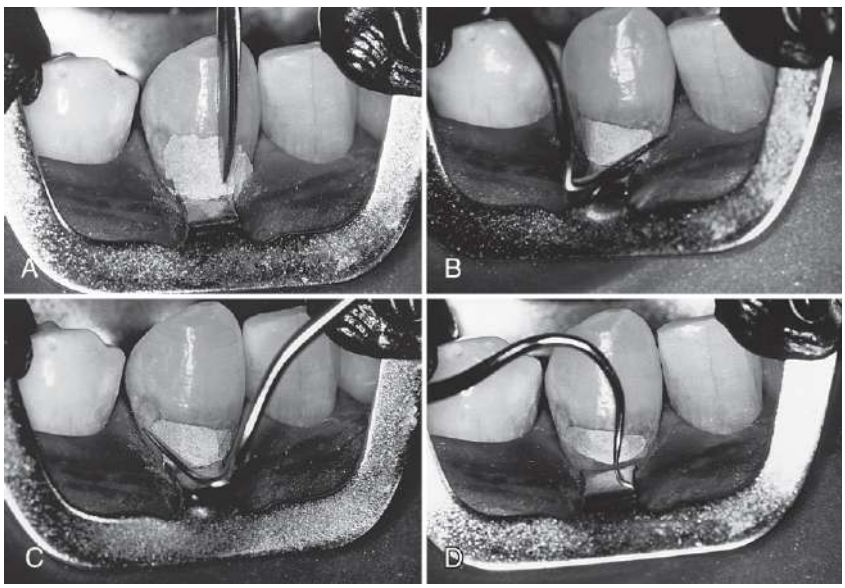


Online Figure 26.38 Inserting amalgam. (A) Place amalgam into the preparation in small increments. (B) Condense first into the retention grooves with a small condenser. (C) Condense against the mesial and distal walls. (D) Overfill and provide sufficient bulk to allow for carving.

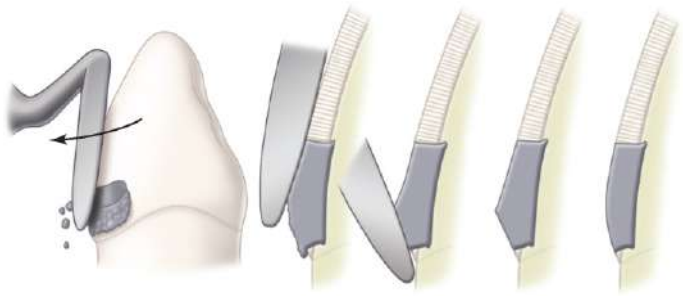


Online Figure 26.39 Use a large condenser or a flat-bladed instrument to offer resistance to condensation pressure applied elsewhere on the restoration.

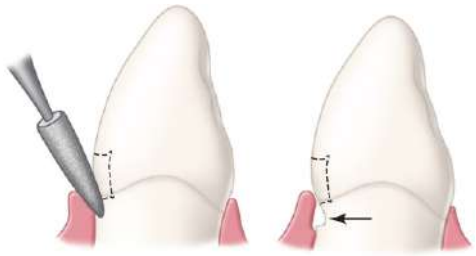
instrument always should rest on the unprepared tooth surface adjacent to the prepared cavosurface margin; this prevents overcarving. The carving procedure is begun by removing excess amalgam to expose the incisal (or occlusal) margin. Removal of excess material continues until the mesial and distal margins are exposed. Finally, material excess at the gingival margin is carved away. Carving the marginal areas should result in developing the desired convex contours in the completed restoration. Improper use of the carving instruments results in a poorly contoured restoration. Note in Online Fig. 26.41 how carving instruments are positioned to provide the desired



Online Figure 26.40 Carving and contouring the restoration. (A) Begin the carving procedure by removing any excess and locating the incisal margin. (B, C) An explorer may be used to remove the excess and locate the mesial and distal margins. (D) Remove the excess and locate the gingival margin.



Online Figure 26.41 Positioning of the carving instrument to prevent overcarving amalgam and to develop the desired gingival contours.



Online Figure 26.42 Incorrect use of a pointed stone at the gingival margin results in the removal of cementum, notching of the tooth structure gingival to the margin, or both.

contours. No amalgam excess should remain at the margins because amalgam may break away, creating a defect at the margin, or cause gingival irritation.

In some instances, it is appropriate to change facial contours because of altered soft tissue levels (e.g. cervical lesions in periodontally treated patients). Facial contours may be increased (or relocated) only enough to prevent food impaction into the gingival sulcus and to provide access for the patient to clean the area. Overcontouring must be avoided because it results in reduced stimulation and cleansing of the gingiva during mastication.

When a rubber dam and the No. 212 cervical retainer have been used for isolation, the retainer is removed using care to open the jaws of the retainer wide enough to prevent marring the surface of the restoration. The rubber dam is removed, and the area is examined carefully to ensure that no amalgam particles remain in the sulcus.

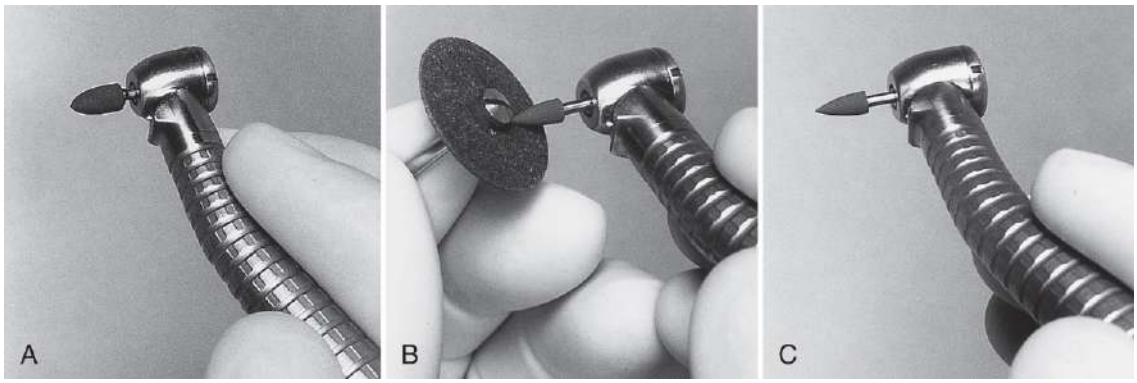
When a retraction cord is used for isolation, it may interfere with carving any excess amalgam at the gingival margin. If so, removal of the gross excess of amalgam is followed by careful removal of the cord, followed by completion of the carving along the margin.

Finishing and Polishing of the Amalgam

If carving procedures were performed correctly, no finishing of the restoration should be required. A slightly moistened cotton pellet held in cotton pliers may be used to further smooth the carved restoration. Additional finishing and polishing of amalgam restorations may be necessary, however, to correct a marginal discrepancy or to improve contour. Care is required when using stones or any rotating cutting instruments on margins positioned below the cementoenamel junction (CEJ) because of the possibility of removing cementum, notching the tooth structure gingival to the margin, or both (Online Fig. 26.42). Online Fig. 26.43 illustrates re-shaping a rubber abrasive point to allow optimal access to the gingival portion of a class V amalgam restoration.

Summary

The clinical usage of amalgam in class III and V situations is not popular as esthetic alternatives with adhesive restorations offer a more clinically acceptable choice for patients.



Online Figure 26.43 Reshaping a rubber abrasive point against a mounted carborundum disk.

Page left intentionally blank

Chapter 27

Additional Information on Instruments and Equipment for Tooth Preparation

"Nothing is impossible; the word itself says I'm possible!"

—Audrey Hepburn

Sharpening Hand Instruments

Selecting the proper hand cutting instrument and using the proper instrument grasp mean little if the instruments are not sharp. Instruments with dull cutting edges cause more pain to the patient, prolong operating time, are more difficult to control and reduce quality and precision in tooth preparation. It is essential that all cutting instruments be sharp. Re-sharpening requires little time and is rewarding. The dentist or the assistant should regularly test the instruments for sharpness, and when indicated, the hand instruments should be sharpened before they are placed in the tray setup, thus preventing delays in starting or completing an operation (see the section on Sharpness Test below).

Many types of sharpening equipment exist, including stationary sharpening stones, mechanical sharpeners and stones that are used in the handpiece. One type or design usually does not accommodate the full variety of dental instruments with their various shapes of cutting edges. For efficient and effective sharpening, the dentist must seek out the most suitable equipment.

Stationary Sharpening Stones

The most frequently used sharpening equipment consists of a block or stick of abrasive material called stone. The stone is supported on a firm surface, and the instrument is oriented and held by hand while being stroked against the stone surface. Stationary stones are often called oilstones because of the common practice of applying a coating of oil on them as an aid to the sharpening process. Sharpening stones are available in a variety of grits, shapes and materials.

Stationary oilstones are available in coarse, medium and fine grits. Only a fine-grit stone is suitable for the final sharpening of dental instruments to be used for tooth preparation. Coarse and medium grits may be used for initial reshaping of a badly damaged instrument or for sharpening other dental equipment such as bench knives. Coarser stones cut more rapidly but produce a rougher surface. If the use of two or more grits is required, the coarser one is used as little as needed for reshaping, and then the final sharpening is done with a fine stone.

Stationary stones can be obtained in various shapes, including flat, grooved, cylindrical and tapered. Flat stones are preferred for sharpening all instruments with straight cutting edges; other shapes are most useful for sharpening instruments with curved cutting edges. Cylindrical stones are used for sharpening instruments with concave edges, and tapered stones permit the use of a portion of the stone with a curvature matching that of the instrument.

Sharpening stones are made from any of several natural or synthetic materials. The normal manufacturing process for the synthetic materials involves pressing carefully sized particles of an abrasive into the desired shape and heating to form a solid. To maintain sharp edges on the particles, the process must result in a porous material. The properties of the stone depend on the volume and size of the pores and on the composition and size of the abrasive. Four types of materials are in common used for sharpening stones: Arkansas stone, silicon carbide (SiC), aluminium oxide and diamond.

Arkansas stone is naturally occurring mineral containing microcrystalline quartz and traditionally has been the preferred material for fine sharpening stones. It is semi-translucent, white or gray in colour and hard enough to sharpen steel, but not carbide instruments. Arkansas stones

are available in hard and soft varieties. The hard stone, although it may cut more slowly, is preferable because the soft stone scratches and grooves easily, rendering it useless. These stones should be lubricated with light machine oil before being used. This assists in the fineness of sharpening, prevents clogging of the stone pores and avoids the creation of heat, which alters the temper of the steel blade. An Arkansas stone should be covered with a thin film of oil when stored. During the sharpening of an instrument, the fine steel cuttings remain on the stone and which tend to fill up the pores of the stone; when the stone appears dirty, it should be wiped with a clean woollen cloth soaked in oil. If the stone is extremely dirty or difficult to clean, it may be wiped with a cloth soaked in alcohol.

SiC is widely used as an industrial abrasive. It is the most commonly used material for grinding wheels and 'sandpapers' and for sharpening stones. It is hard enough to cut steel effectively, but not hard enough to sharpen carbide instruments. SiC stones are available in many shapes in coarse and medium grits, but not in fine grits. As a result, they are not as suitable as other materials for the final sharpening of dental instruments. SiC stones are normally of a dark colour, often black or greenish black. These stones are moderately porous and require lubrication with a light oil to prevent clogging.

Aluminium oxide is increasingly used to manufacture sharpening stones. Aluminium oxide stones commonly are produced in various textures from different particle sizes of abrasive. Coarse and medium grit stones generally appear as speckled tan or brownish in colour. Fine-grit stones are usually white, have superior properties and are less porous so that they require less lubrication during use. Either water or light oil is adequate as a lubricant.

Diamond is the hardest available abrasive and is most effective for cutting and shaping hard materials. It is the only material routinely capable of sharpening carbide and steel instruments. Diamond hones are small blocks of metal with fine diamond particles impregnated in the surface. The diamonds are held in place by an electroplated layer of corrosion-resistant metal. Most hones include grooved and rounded surfaces and a straight surface and are adaptable for sharpening instruments with curved blades. These hones are non-porous, but the use of a lubricant extends the life of the hones. They may be cleaned with a mild detergent and a medium-bristle brush.

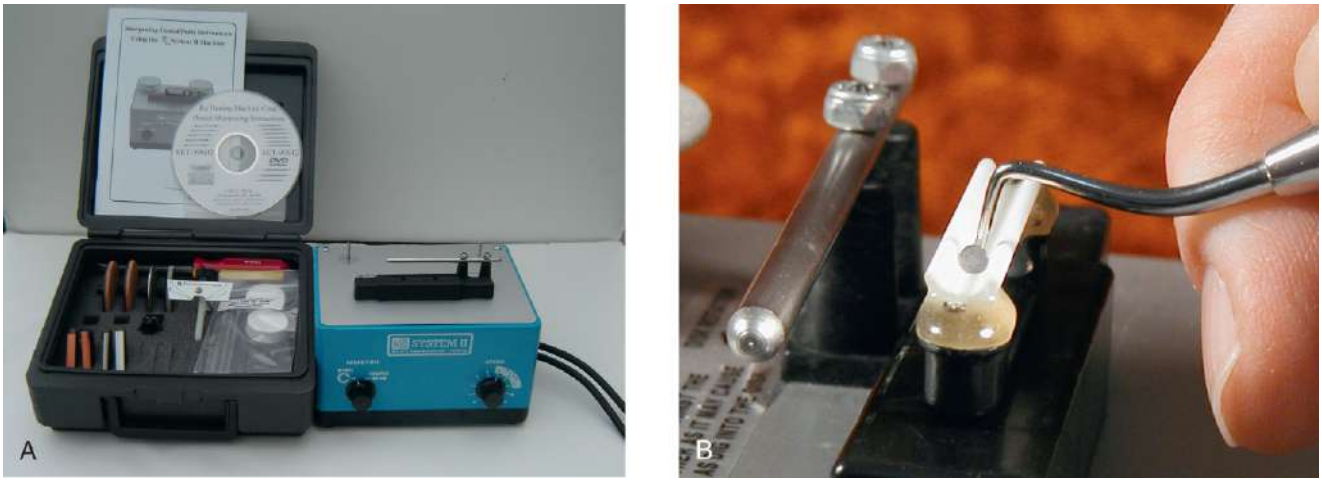
Mechanical Sharpeners

As high-speed rotary cutting instruments have been improved and their use has increased, the use of hand cutting instruments and the need for re-sharpening has decreased. As a result, some dental office personnel do not do enough hand sharpening to remain confident of their proficiency. Under such circumstances, the use of a powered mechanical sharpener is beneficial.

The Rx Honing Machine (Rx Honing Machine Corp, Mishawaka, IN, USA) is an example of a mechanical sharpener (Online Fig. 27.1). This instrument moves a hone in a reciprocating motion at a slow speed, while the instrument is held at the appropriate angulation and supported by a rest. This is much easier than holding the instrument at the proper angulation while moving it relative to the hone. Interchangeable aluminium oxide hones of different shapes and coarseness are available to accommodate the various instrument sizes, shapes and degrees of dullness. Restoration of the cutting edge is accomplished more easily and in less time than by other sharpening methods. This type of sharpener is also very versatile and, with available accessories, can fill almost all instrument sharpening needs.

Handpiece Sharpening Stones

Mounted SiC and aluminium oxide stones for use with straight and angle handpieces are available in various



Online Figure 27.1 (A) The Rx Honing Machine shown is used as mechanical sharpener for many different types of dental instruments. It has two spindle drives, one clockwise and the other counterclockwise, to which can be mounted different types of disks and hones to polish and sharpen various types of dental instruments. (B) The RX Honing Machine shown using a ceramic hone to sharpen a Black spoon dental instrument to extend the life of the instrument.

sizes and shapes. Those intended for use in straight handpieces, particularly the cylindrical instruments with straight-sided silhouettes, are more useful for sharpening hand instruments than are the smaller points intended for intra-oral use in the angle handpieces. Because of their curved periphery, it is difficult to produce a flat surface using any of these instruments. These stones also may produce inconsistent results because of the speed variables and the usual lack of a rest or guide for the instrument. Satisfactory results can be obtained, however, with minimal practice, especially on instruments with curved blades.

Principles of Sharpening

Most operative hand cutting instruments can be sharpened successfully on either a stationary stone or the mechanical sharpener. The secret to easy and successful sharpening is to sharpen the instrument at the first sign of dullness and not wait until the edge is completely lost. If this procedure is followed, a fine cutting edge is restored with a few strokes on a stationary stone or a light touch to the mechanical sharpener. At the same time, operating efficiency is not reduced by attempting to use an instrument that is getting progressively dull.

The choice of equipment used for sharpening is up to the dentist. In the use of any equipment, several basic principles of sharpening should be followed:

I. Sharpen instruments only after they have been cleaned and sterilized.
II. Establish the proper bevel angle (usually 45 degrees) and the desired angle of the cutting edge to the blade before placing the instrument against the stone, and maintain these angles while sharpening.
III. Use a light stroke or pressure against the stone to minimize frictional heat.
IV. Use a rest or guide, whenever possible.
V. Remove as little metal from the blade as possible.
VI. Lightly hone the un-bevelled side of the blade after sharpening, to remove the fine bur that may have been created.
VII. After sharpening, re-sterilize the instrument along with other items on the instrument tray setup.
VIII. Keep the sharpening stones clean and free of metal cuttings.

Mechanical Sharpening Techniques

When chisels, hatchets, hoes, angle formers or gingival margin trimmers are sharpened on a reciprocating honing machine (i.e. sharpener), the blade is placed against the steady rest, and the proper angle of the cutting edge of the blade is established before starting the motor. Light pressure of the instrument against the reciprocating hone is maintained with a firm grasp on the instrument. A trace of metal debris on the face of a flat hone along the length of the cutting edge is an indication that the entire cutting edge is contacting the hone (Fig. 27.1B).

The mechanical sharpener is easily mastered with a little practice and is a quick method of sharpening hand instruments. Regardless of the type of mechanical sharpener used, the associated instructions for use should be thoroughly understood before attempting to sharpen any type of instrument.

Handpiece stones are used chiefly for instruments with curved blades, especially for the inside curve of such blades. The handpiece should be run at a low speed. The instrument is held lightly against the stone with a modified pen grasp, and whenever possible, the ring and little fingers of each hand should be touching each other to act as a rest or steadying force. When this method of sharpening is used, care must be exercised not to overheat the instrument being sharpened. The use of some form of lubricant or coolant is advisable. If oil is used, care should be exercised to ensure that oil is not thrown from the stone during sharpening, and the stone should be reserved for future sharpening only.

An instrument such as an amalgam knife or a gold knife has a wide blade with a narrow edge bevel, in contrast to the wide bevel of a chisel or hatchet. It is difficult to maintain the narrow edge bevel by using a mechanical sharpener or a handpiece stone. This type of instrument should be sharpened on a stationary stone.

Stationary Stone Sharpening Techniques

The stationary sharpening stone should be at least 2-in. wide and 5-in. long because a smaller stone is impractical. It also should be of medium grit for hand cutting instruments. Before the stone is used, a thin film of light oil should be placed on the working surface. In addition to establishing the proper 45-degree angle of the bevel and the cutting edge to the stone, several fundamental rules apply to using the stationary stone:

I. Lay the stone on a flat surface, and do not tilt the stone while sharpening.
II. Grasp the instrument firmly, usually with a modified pen grasp, so that it does not rotate or change angles while being sharpened.
III. To ensure stability during the sharpening strokes, use the ring and little finger as a rest, and guide along a flat surface or along the stone. This prevents rolling or dipping of the instrument, which results in a distorted and uneven bevel.
IV. Use a light stroke to prevent the creation of heat and the scratching of the stone.
V. Use different areas of the stone's surface while sharpening because this helps prevent the formation of grooves on the stone that impair efficiency and accuracy of the sharpening procedure.

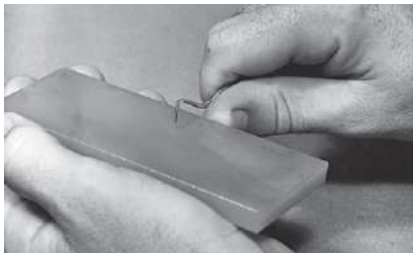
When sharpening chisels, hatchets or hoes on the stationary stone, grasp the instrument with a modified pen grasp,



Online Figure 27.2 Sharpening an instrument. Maintaining the proper angle of bevel and angle of the cutting edge to the stone is aided by resting the fingertips on the stone.

place the blade perpendicular to the stone and tilt the instrument to establish the correct bevel (Online Fig. 27.2). Establishing and maintaining this correct bevel is the most difficult part of sharpening on a stationary stone. One method that assists in establishing the proper bevel angle is to observe the oil on the stone while the instrument is tilted, in an effort to establish contact between the entire bevel and the stone. When oil is expressed evenly on all sides, the entire bevel is touching the stone, and the proper angle has been established to proceed with the sharpening strokes. If this alignment is altered during sharpening, discrepancies of the cutting edge and bevel result. Using the finger rests and guides as illustrated in Online Figure 27.2, the operator can slide the instrument back and forth along the stone. The motivating force should be from the shoulder so that the relationship of the hand to the plane of the stone is not changed during the stroke. Another technique is to move the stone back and forth while maintaining a constant position of the instrument. The procedure for sharpening angle formers is essentially the same as that used for chisels, hatchets or hoes except that allowance must be made for the angle of the cutting edge to the blade.

Gingival margin trimmers require more orientation of the cutting edge to the stone before sharpening than does a regular hatchet. The same principle of establishing the proper bevel angle and cutting edge angle is the criterion for instrument position before sharpening. It may be expedient to use a palm-and-thumb grasp when sharpening a trimmer with a 95- or 100-degree cutting edge angle (Online Fig. 27.3).

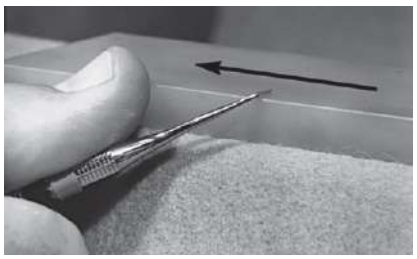


Online Figure 27.3 Sharpening the gingival margin trimmer. The palm-and-thumb grasp may be used while holding the stone in the opposite hand to establish a proper cutting edge angle.

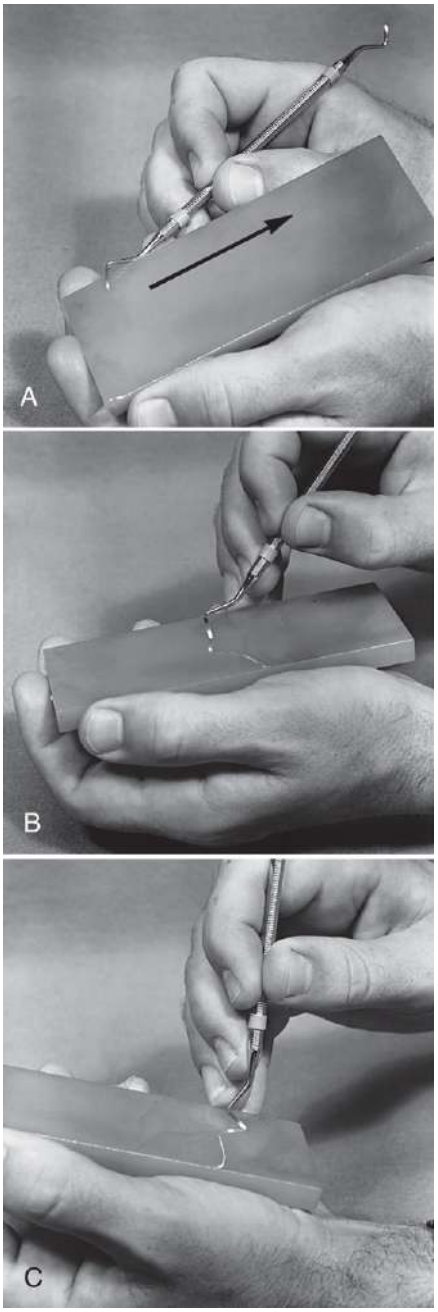
When single-bevel instruments are sharpened, a thin, rough ridge of distorted metal, called bur or bur-edge collects on the un-bevelled side of the blade. This bur is eliminated by a light stroke of the un-bevelled side of the blade over the stone. This side of the blade is placed flat on the stone, and one short forward stroke is made. Burring can be kept to a minimum, however, if the direction of the sharpening stroke is against only the cutting edge of the blade, and the cutting edge does not contact the stone during the return stroke. The blade is touching the stone only on the forward sharpening stroke.

The amalgam knife or the gold knife has a thin blade tapering to the sharpened edge. A narrow edge bevel is present on both sides of the blade. In sharpening this instrument, only the edge bevels should be honed. If the entire side of the blade is worked each time, the thin blade soon disappears or becomes so thin that it fractures under the slightest pressure. To sharpen the amalgam knife or the gold knife, the blade is placed on the stone with the junction of the blade and shank immediately over the edge of the stone. The blade is tilted to form a small acute angle with the surface of the stone, and the stroke is straight along the stone and towards the edge of the blade only (Online Fig. 27.4). The sharpening is accomplished on both sides of the blade, with the stroke always towards the blade edge. This method produces the finest edge and eliminates any burs on the cutting edge.

The most difficult instruments to sharpen on a flat stone are the spoon excavators and discoid instruments. Only the rounded outside surface of the spoon can be honed satisfactorily on a flat stone, and this involves a rotary movement accompanied by a pull stroke to maintain the curvature of the edge. The spoon is placed on the far end of the stone and held so that the handle is pointing towards the operator. As the instrument is pulled along the stone towards the operator, the handle is rotated gradually away from the operator, until it is pointing away from the operator at the end of the stroke. The instrument is picked up and placed at the far end of the stone, and the motion is repeated until the edge is honed. The stone may be placed on a flat surface or held in the hand for this procedure (Online Fig. 27.5). To hone the flat inside surface of the blade, a small cylindrical stone is passed back and forth over the surface (Online Fig. 27.6).



Online Figure 27.4 Sharpening an amalgam knife or a gold knife. The stone is placed at the edge of the table so that the blade may be tilted to form an acute angle with the stone. The arrow indicates the direction of the sharpening movement of the instrument along the stone.

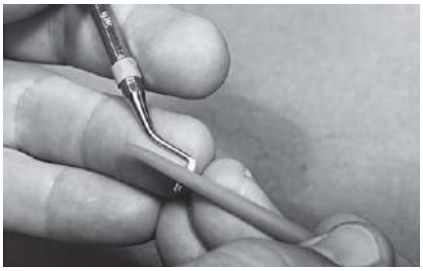


Online Figure 27.5 Sharpening a spoon excavator. (A) Beginning of the stroke. (B) Continuation of the pull stroke while rotating the handle in a direction opposite the stroke. (C) Completion of the stroke and the handle rotation. Finger guides are used during the entire stroke, which is in the direction indicated by the arrow.

Other means of sharpening spoon excavators are achieved by using a grooved stone, mounted disks or stones for use with a straight handpiece. A tendency exists, however, to remove too much metal when handpiece stones are used.

Sharpness Test

Sharpness of an instrument can be tested by lightly resting the cutting edge on a hard plastic surface. If the cutting edge digs in during an attempt to slide the instrument forward over the surface, the instrument is sharp. If it slides, the instrument is dull. Only very light pressure is exerted in testing for sharpness.



Online Figure 27.6 Use of a small cylindrical stone to hone the inside surface of spoon excavators and discoid–cleoid instruments. *Note*: Gloving is not illustrated in Figures 7–15 to 7–19 because sharpening is accomplished *after* sterilizing the washed instrument; after the instrument is sharpened, it is sterilized again.

The principles and techniques previously discussed provide sufficient background for the operator to use proper methods in sharpening other instruments not discussed here. It cannot be overemphasized that sharp instruments are necessary for optimal operating procedures. It also has been found prudent to have multiple tray setups so that a substitute instrument is available, if necessary, or substitute sterile instruments should be available so that other sterile tray setups are not disrupted when instruments are borrowed.

Sterilization and Storage of Hand Cutting Instruments

Because hepatitis A and B viruses have been found in the saliva of infected persons, and evidence indicates that some dental personnel have acquired hepatitis B infections from patients, the importance of proper equipment and procedures for instrument sterilization must be emphasized. Sterilization in dental offices can be accomplished by autoclaving, dry-heat procedures, ethylene oxide equipment and chemical vapour sterilizers. Boiling and chemical solutions (cold disinfection) do not sterilize instruments and should be considered as disinfection procedures only. The belief that only instruments that puncture or cut soft tissue or are exposed to blood should be sterilized and others only disinfected is no longer valid as a precaution against cross-infections. Storage of any hand cutting instrument should be in a sterile, wrapped tray setup or in an individual sterile wrapping.

Effects of Sterilization

Methods of sterilization are sporicidal cold disinfection, boiling in water, autoclaving (steaming under pressure), use of chemical vapour and use of hot air (dry heat) (Online Chapter 19 or details regarding acceptable

methods of sterilization). Sterilizing carbon steel instruments by any of the first three methods causes discoloration, rust and corrosion. Several methods for protecting against or minimizing these problems are available. One method used by manufacturers is to electroplate the instrument. This affords protection, except on the blade, where use and sharpening remove the plating. The plating also may pit or peel on the handle and shank under certain circumstances. A second method of protection is the use of rust inhibitors, which are soluble alkaline compounds. These usually are incorporated into commercial sporicidal cold disinfectant solutions, and special preparations are available for use in boiling water and autoclaves. The third method of minimizing the effect of moisture is to remove the instruments promptly at the end of the recommended sterilizing period, dry them thoroughly, and place them in the instrument cabinet or on the tray setup. Leaving instruments exposed to moisture for extended periods or overnight should be avoided.

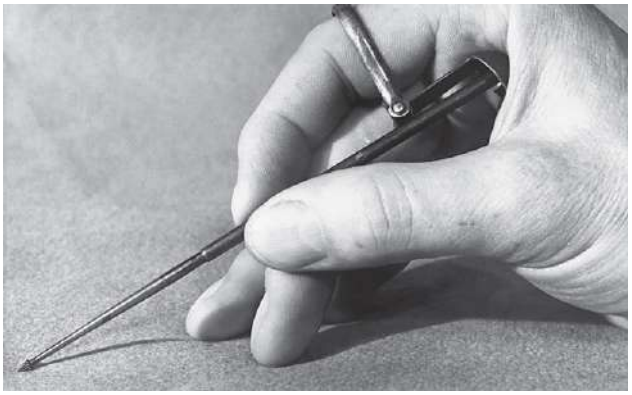
Boiling in water and autoclaving for sterilization does not produce discoloration, rust or corrosion of stainless steel instruments. Prolonged immersion in cold disinfectant solutions may, however, cause rusting. It is advisable to leave stainless steel instruments exposed to moisture only for the recommended time. Dry-heat sterilizers do not rust and corrode carbon steel instruments, but the high heat may reduce the hardness of the alloy, which would reduce the ability of the instruments to retain their sharp cutting edge. The choice of alloy in a hand instrument is left to the operator, but whichever alloy is selected to suit the immediate needs would soon prove unsatisfactory if proper manipulation and sterilization are not continually practiced.

Powered Cutting Equipment

Development of Rotary Equipment

The availability of some method of cutting and shaping of tooth structure and restorative materials is essential for the restoration of teeth. Although archaeological evidence of dental treatment dates from 5000 BC, little is known about the equipment and methods used then.[1] Early drills powered by hand are illustrated in Online Figs. 27.7 and 27.8. Much of the subsequent development leading to present powered cutting equipment can be seen as a search for improved sources of energy and means for holding and controlling the cutting instrument. This search has culminated in the use of replaceable bladed or abrasive instruments held in a rotary handpiece, usually powered by compressed air.

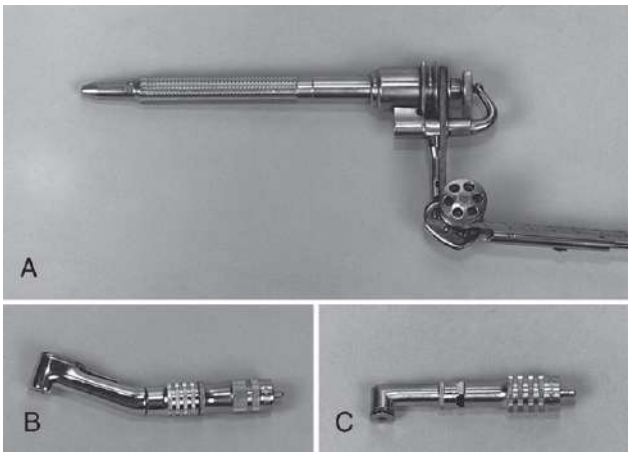
A handpiece is a device for holding rotating instruments, transmitting power to them and for positioning them intra-orally. Handpieces and associated cutting and polishing instruments developed as two basic types, straight and angle (Online Fig. 27.9). Most of the development of methods for preparing teeth has occurred within the last 100 years.[2] Effective equipment for the removal (or preparation) of enamel has been available only since 1947, when speeds of 10,000 rpm were first used, along with newly marketed carbide burs and diamond instruments.



Online Figure 27.7 Early straight hand drill for direct access preparations (circa 1800). The back end of the bur shank fits into a finger ring while the front end is rotated with the thumb and the forefinger.



Online Figure 27.8 Early angle hand drill for indirect access preparations (circa 1850). The bur is activated by squeezing the spring-loaded handle.



Online Figure 27.9 Conventional designs of handpieces. (A) Belt-driven straight handpiece. (B) Gear-driven angle handpiece that attaches to front end of the straight handpiece. (C) Gear-driven angle handpiece designed for cleaning and polishing procedures.

Online Table 27.1 Evolution of rotary cutting equipment in dentistry

Date	Instrument	Speed (rpm)
1728	Hand-rotated instruments	300
1871	Foot engine	700
1874	Electric engine	1000
1914	Dental unit	5000
1942	Diamond cutting instruments	5000
1946	Old units converted to increase speed	10,000
1947	Tungsten carbide burs	12,000
1953	Ball-bearing handpieces	25,000
1955	Water-turbine angle handpiece	50,000
1955	Belt-driven angle handpiece (Page-Chayes)	150,000
1957	Air-turbine angle handpiece	250,000
1961	Air-turbine straight handpiece	25,000
1962	Experimental air-bearing handpiece	800,000
1994	Contemporary air-turbine handpiece	300,000

rpm, revolutions per minute.

Since 1953, continued improvements in the design and materials of construction for handpieces and instruments have resulted in equipment that is efficient and sterilizable, much to the credit of manufacturers and the profession alike. Online Table 27.1 summarizes some significant developments of rotary dental equipment.

One of the most significant advancements was the introduction of the electric motor as a power source in 1874. It was incorporated into a dental unit in 1914.[3] The initial handpiece equipment and operating speeds (maximum of 5000 rpm) remained virtually unchanged until 1946 (Online Fig. 27.10). The steel burs used at the time could not cut enamel effectively, even when applied with a great force. With steel burs, increased speed and power resulted only in increased heat and instrument wear. Further progress was delayed until the development of instruments that could cut enamel. Diamond cutting instruments were developed in Germany around 1935 but were scarce in the United States until after World War II. In a 10-year period, starting in late 1946, cutting techniques were revolutionized. Diamond instruments and tungsten carbide burs capable of cutting enamel were produced commercially. Both instruments performed best at the highest speeds available, and that prompted the development of higher speed handpieces. Obtaining speeds of 10,000 to 15,000 rpm was a relatively simple matter of modifying existing equipment by enlarging the drive pulleys on the dental engine. By 1950, speeds of 60,000 rpm and greater had been attained by newly designed equipment employing speed-multiplying internal belt drives (Online Fig. 27.11).[2] They were found to be more effective for cutting tooth structure and for reducing perceived vibration.

The major breakthrough in the development of high-speed rotary equipment came with the introduction of



Online Figure 27.10 Typical equipment when an electric motor is used as the source of power: foot control with rheostat (*w*), belt-driven straight handpiece (*x*), three-piece adjustable extension arm (*y*) and electric motor (*z*).



Online Figure 27.11 Page-Chayes handpiece (circa 1955). The first belt-driven angle handpiece to operate successfully at speeds greater than 100,000 rpm.



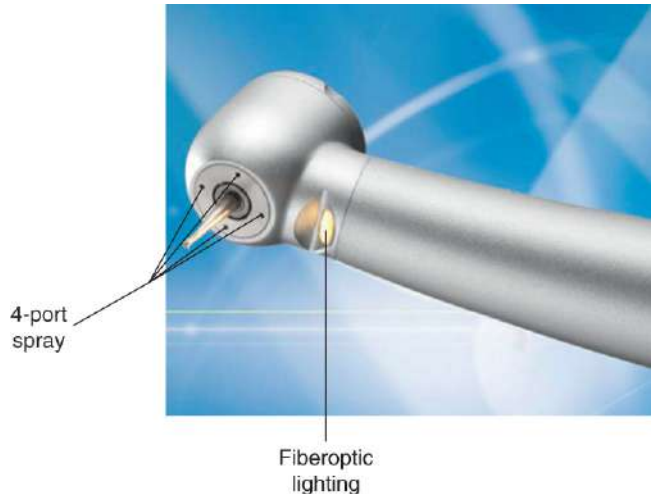
Online Figure 27.12 Turbo-Jet portable unit (circa 1955). A small turbine in the head of the angle handpiece is driven by water circulated by a pump housed in the mobile base.



Online Figure 27.13 Air-turbine handpiece. (A) The Borden Airotor handpiece (circa 1957) was the first clinically successful air-turbine handpiece. Current air-driven handpieces are similar in the basic design. (B) Air-turbine straight handpiece (circa 1980).



Online Figure 27.14 Contemporary contra-angle air-turbine handpiece connected to the air-water supply line.
(Courtesy: KaVo Dental Corp., Charlotte, NC, USA).



Online Figure 27.15 View of the handpiece showing four spray ports for cooling and fibre-optic illumination.
(Courtesy: KaVo Dental Corp., Charlotte, NC, USA)

contra-angled handpieces with internal turbine drives in the contra-angle head.[4] Early units were water driven, but subsequent units were air driven (Online Figs. 27.12 and 27.13A). Although most current air-turbine handpieces (Online Fig. 27.14) have free-running speeds of approximately 300,000 rpm, the small size of the turbine in the head limits their power output. The speed can decrease to 200,000 rpm or less, with small lateral workloads during cutting, and the handpiece may stall at moderate loads.[5] This tendency to stall under high loads is an excellent safety feature for tooth preparation because excessive pressure cannot be applied. Air-driven handpieces continue to be the most popular type of handpiece equipment because of the overall simplicity of design, ease of control, versatility and patient acceptance. The external appearance of current handpieces is similar to the earliest models.

The low torque and power output of the contra-angle turbines made them unsuitable for some finishing and polishing techniques, for which large heavy instruments are needed. The application of the turbine principle to the straight handpiece eliminated the necessity of having an electric engine as part of a standard dental unit. The design of the straight handpiece turbine provided the desirable high torque for low-speed operation (Fig. 27.13B).

Increasing concern about patient-to-patient transfer of infectious agents has put emphasis on other aspects of handpiece performance. Advancements in straight and angle handpieces allow repeated sterilization by several methods (Online Chapter 19). Sterilization produces some damage



Online Figure 27.16 Electric handpieces and unit.
(Courtesy: DENTSPLY International, York, PA, USA)

to parts of the handpiece, however, necessitating more frequent service and repair. Other improvements of the angle handpiece include smaller head sizes, more torque, lower noise levels and better chucking mechanisms. Since 1955, angle handpieces have had an air-water spray feature to provide cooling, cleansing and improved visibility.[6] Most modern-angled handpieces also include fibre-optic lighting of the cutting site (Online Fig. 27.15). Electric handpieces that compete effectively with air-turbine designs have also been developed (Online Fig. 27.16).

Summary

A thorough knowledge regarding information on instruments and equipment for tooth preparation is important to make the clinical procedure effective and efficient.

References

1. Guerini V: *A history of dentistry*, Philadelphia, 1909, Lea & Febiger.
2. Sockwell CL: Dental handpieces and rotary cutting instruments, *Dent Clin North Am* 15:219–244, 1971.
3. SS White Dental Manufacturing Company: *A century of service to dentistry*, Philadelphia, 1944, SS White Dental Manufacturing.
4. Nelson RJ, Pelander CE, Kumpala JW: Hydraulic turbine contra-angle handpiece, *J Am Dent Assoc* 47:324–329, 1953.
5. Taylor DF, Perkins RR, Kumpala JW: Characteristics of some air turbine handpieces, *J Am Dent Assoc* 64:794–805, 1962.
6. Peyton FA: Effectiveness of water coolants with rotary cutting instruments, *J Am Dent Assoc* 56:664–675, 1958.

Page left intentionally blank